

Switching formation shape control with distance + area/angle feedback[☆]

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ABSTRACT

A well known problem with distance-based formation control is the existence of multiple equilibrium points not associated with the desired formation. This problem can be mitigated by introducing an additional controlled variable. In this paper, we consider distance + area (or angle) schemes for 2D formations of single-integrator agents. By using directed graphs and triangulation of the n -agent formation, we introduce switching control laws that ensure the asymptotic stability of the desired formation for almost all initial agent positions. The state-dependent switching strategy is designed to force the formation to escape the subspaces of the incorrect equilibria. The switching strategy also allows us to remove restrictions on the shape of the desired formation that were present in previous results.

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1. Introduction

In collaborative robotic tasks (e.g., perimeter protection, co-transportation of large objects, search and rescue), it is often required that the robots acquire and maintain a desired spatial pattern. This type of multi-agent coordination problem is commonly referred to as formation shape control.

Three methods are typically used in formation shape control: (a) regulate the *global position* of each agent [1,2]; (b) regulate the *relative position* of agent pairs [3,4]; or (c) regulate a set of inter-agent *distances* (magnitude of the relative position vector) [5,6]. The first method requires an external sensor such as an overhead camera, a Vicon-type motion system, or GPS. Thus, this method is centralized and has a single point of failure. The second method requires the agents to have a common global coordinate frame or local coordinate frames with the same orientation, which may not be feasible in practice. Unlike the previous two methods, the feedback variables in the third method (a.k.a. distance-based control) can be computed with respect to the agent's local frame which does not need to have a common orientation. Therefore, the agents can converge to any translated and/or rotated version of the desired formation.

A problem with the distance-based approach is how to prevent agents from converging to a formation that has a different shape or orientation as the desired formation but that satisfies the set

of distance constraints. That is, since the agents' relative positions are not uniquely defined by the distance constraints, ambiguities can occur in the positioning of the agents [7]. Through rigid graph theory, one can partially solve this problem since it prescribes the minimal number of distances to be controlled, thereby reducing the ambiguous formations to ones that are just flipped versions of the desired formation [7]. In such cases, the agents' initial positions determine if the system converges to the desired formation or to an ambiguous one. This means that rigidity-based distance-only formation controllers have non-global stability properties.

Variants of the distance-based approach have been recently proposed to deal with the aforementioned convergence problem. In [8], an approach that combines distance and signed area constraints was introduced to prevent flipped formations in 2D, ensuring that the target formation is uniquely acquired. However, the method was only analyzed for special formations of three and four agents. Recently in [9], we extended the distance + signed area approach to directed formations of n agents that meet some shape and control gain conditions. A related method [10] uses the distance and the angle between agents to decrease the likelihood of the system converging to ambiguous planar formations. The stability of the control in [10] is still local although it is claimed that increasing a control gain would expand its region of attraction.

A common issue with the above distance + area/angle results is that although the controller avoids some undesired equilibrium points due to the additional controlled variable, they introduce new ones since the potential function has multiple local minima. This was partially mitigated by imposing conditions on the control gains and/or shape of the desired formation. In this

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paper, we seek to remove some of the restrictions present in the previous distance + area/angle results. Specifically, in comparison to [9], we remove all conditions on the shape of the desired formation. Moreover, our region of attraction includes almost all initial agent positions whereas this region was undefined in [10]. This improvement is enabled by a state-dependent switching control scheme. The proposed switching logic is inspired by the observation that the distance-only controller for triangular formations has three invariant subspaces [11]: (1) the subspace where the three agents are positioned collinearly, (2) a non-collinear subspace that includes the desired formation, and (3) another non-collinear subspace that includes a flipped version of the desired formation. The non-collinear subspaces are disconnected and separated by the collinear one. Our switching logic is based on the fact that the signed area or the angles assume different values in the two non-collinear subspaces. In particular, we use a distance + area (or angle) controller when the actual formation is in the incorrect non-collinear subspace or in the collinear subspace, and the standard distance-only controller when the formation is in the correct non-collinear subspace. That is, the additional controlled variable (area or angle) allows the formation to escape the undesired subspaces. This strategy is applied to systems with any number of agents by triangulating the formation such that the formation graph is a leader-first-follower type of minimally persistent directed graph [12]. The stability of our closed-loop system is analyzed using switched system theory [13]. We prove the asymptotic stability of the desired formation as long as the leader and first follower in the distance + area scheme (or any two agents connected with a graph edge in the distance + angle scheme) are not collocated at time zero.

2. Background

2.1. Undirected graphs

A framework F in $\mathbb{R}^2$ is defined by an undirected graph G of $N \geq 3$ vertices with coordinates $p_i \in \mathbb{R}^2$, $i = 1, \dots, N$ and $a \in \{1, \dots, N(N-1)/2\}$ undirected edges. For convenience, we will write $F = (G, p)$ where $p = [p_1, \dots, p_N] \in \mathbb{R}^{2N}$.

The edge function $\psi : \mathbb{R}^{2N} \rightarrow \mathbb{R}^a$ of a framework is defined as $\psi(p) = [\dots, \|p_i - p_j\|^2, \dots]$ where (i, j) is the m th edge in G connecting vertices i and j . Frameworks (G, p) and $(G, \hat{p})$ are said to be equivalent if $\psi(p) = \psi(\hat{p})$ and congruent if $\|p_i - p_j\| = \|\hat{p}_i - \hat{p}_j\|$ for all distinct vertices i and j . From the edge function, one can determine the rigidity matrix $R : \mathbb{R}^{2N} \rightarrow \mathbb{R}^{a \times 2N}$ as follows

$$R(p) = \frac{1}{2} \frac{\partial \psi(p)}{\partial p} \quad (1)$$

where $\text{rank}[R(p)] \leq 2N - 3$ for all p [14]. The rigidity matrix plays an important role in determining the *rigidity* of a framework.

Let the map $\mathcal{T} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be such that $\mathcal{T}(x) = \mathcal{R}x + d$ where $\mathcal{R} \in SO(2)$ and $d \in \mathbb{R}^2$. A framework $F = (G, p)$ is rigid in $\mathbb{R}^2$ if all of its continuous motions satisfy $p_i(t) = \mathcal{T}(p_i)$ for all $i = 1, \dots, N$ and $\forall t \geq 0$ [14,15]. Given the difficulty in characterizing the concept of rigidity in the most general case, we often consider the case of *generic rigidity* [16] where algebraic dependencies among the vertex positions that lead to geometric singularities are avoided. In this case, rigidity is equivalent to the related concept of infinitesimal rigidity which is easier to characterize. Specifically, a framework (G, p) is infinitesimally rigid if and only if $\text{rank}[R(p)] = 2N - 3$ [15].

A rigid framework is minimally rigid if and only if $a = 2N - 3$ [7]. If rigid frameworks (G, p) and $(G, \hat{p})$ are equivalent but not congruent, they are called *ambiguous* [7]. Notice that in such

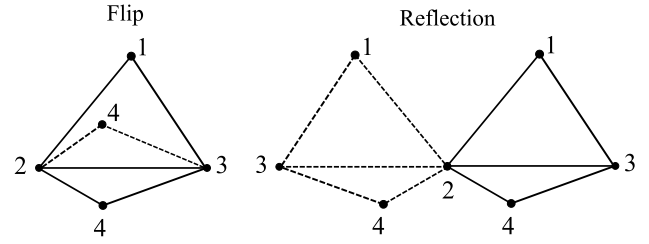


Fig. 1. Types of ambiguous frameworks.

cases the edge function alone does not uniquely characterize the framework. Some familiar ambiguities are depicted in Fig. 1. It is not difficult to see that reflections are the only type of flip ambiguity that can happen in frameworks where $N = 3$ (triangular frameworks).

2.2. Directed graphs

A directed graph G is one where the edge (i, j) is directed, i.e., i is the source vertex of the edge and j is the sink vertex. For convenience, we will write $G = (\mathcal{V}, E)$ where $\mathcal{V} = \{1, \dots, N\}$ is the vertex set and E is directed edge set. The out-degree of $i \in \mathcal{V}$ ($\text{out}(i)$) refers to the number of edges whose source is vertex i and sinks are in $\mathcal{V} - \{i\}$.

The notion of rigidity of directed frameworks requires some additional concepts beyond the ones needed for undirected frameworks. The first one is called *constraint consistency*, which means that every agent (vertex) is capable of meeting all its distance constraints as all the others are trying to do the same [17]. As shown in Lemma 5 of [18], a sufficient condition for a directed graph to be constraint consistent in $\mathbb{R}^2$ is $\text{out}(i) \leq 2$ for all $i \in \mathcal{V}$. The second concept is called *graph persistency*, which means that the structure of the framework is preserved as all agents attempt to satisfy their distance constraints [17]. As shown in Theorem 3 of [18], a persistent directed framework is equivalent to constraint consistency plus the underlying undirected graph being infinitesimally rigid. A persistent framework in $\mathbb{R}^2$ where removing any single edge causes it to lose persistency is called minimally persistent. A sufficient condition for minimally persistency is minimal rigidity and a necessary condition is $\text{out}(i) \leq 2$ for all $i \in \mathcal{V}$ [18]. A minimally persistent framework can be constructed starting from two vertices with an edge by the so-called Henneberg insertion of type I [19]. The method entails iteratively adding a vertex with two outgoing edges. A similar construction procedure applies to rigid (undirected) frameworks by adding undirected edges. We will call a framework constructed as described a Henneberg framework.

2.3. Signed area

The *signed area* of a triangular framework, $S : \mathbb{R}^6 \rightarrow \mathbb{R}$, is given by [8]

$$S(p) = \frac{1}{2} \det \begin{bmatrix} 1 & 1 & 1 \\ p_1 & p_2 & p_3 \end{bmatrix} = \frac{1}{2} (p_3 - p_1)^T J (p_3 - p_2) \quad (2)$$

where $p = [p_1, p_2, p_3]$ and

$$J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \quad (3)$$

The signed area is positive if the vertices are ordered counter-clockwise, negative if ordered clockwise, and zero if any two vertices are collocated or the three vertices are collinear.

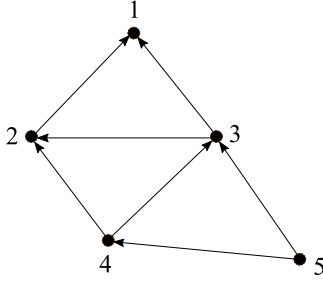


Fig. 2. Signed area example.

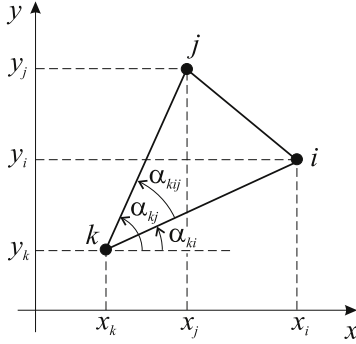


Fig. 3. Definition of the edge angle.

Consider a Henneberg framework with N vertices and directed edge set E^d . After dividing this framework into triangular sub-frameworks, its signed area, $\chi : \mathbb{R}^{2N} \rightarrow \mathbb{R}^{N-2}$, is defined as

$$\chi(p) = \left[\dots, \frac{1}{2} \det \begin{bmatrix} 1 & 1 & 1 \\ p_i & p_j & p_k \end{bmatrix}, \dots \right], \quad \forall (k, i), (k, j) \in E^d - \{(2, 1)\} \quad (4)$$

such that its m th component is related to the signed area of the m th triangle constructed with vertices i, j , and k . As an example, the signed area of the framework in Fig. 2 is

$$\chi(p) = \left[\frac{1}{2} (p_3 - p_1)^T J (p_3 - p_2), \frac{1}{2} (p_4 - p_2)^T J (p_4 - p_3), \frac{1}{2} (p_5 - p_3)^T J (p_5 - p_4) \right], \quad (5)$$

where the three elements are positive, negative, and positive, respectively.

2.4. Edge angle

Given a triangular framework with vertices i, j, k , we introduce the so-called *edge angle* subtended by edges (k, i) and (k, j) , $\alpha_{kij} : \mathbb{R}^6 \rightarrow [0, 2\pi]$:

$$\alpha_{kij}(p) = \text{mod} \{ (\alpha_{kj} - \alpha_{ki}), 2\pi \} \quad (6)$$

where

$$\alpha_{ki} = \text{atan2}(y_i - y_k, x_i - x_k), \quad \alpha_{kj} = \text{atan2}(y_j - y_k, x_j - x_k), \quad (7)$$

$p_l = [x_l, y_l]$, $l = i, j, k$, and $p = [p_k, p_i, p_j]$ (see Fig. 3).

The edge angle of a triangulated Henneberg framework with N vertices and directed edge set E^d , $A : \mathbb{R}^{2N} \rightarrow [0, 2\pi]^{N-2}$, is defined as

$$A(p) = [\dots, \alpha_{kij}, \dots], \quad \forall (k, i), (k, j) \in E^d - \{(2, 1)\} \quad (8)$$

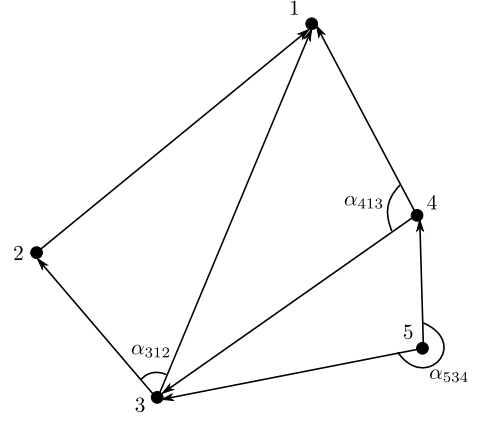


Fig. 4. Edge angle example.

such that its m th component is related to the edge angle of the m th triangular sub-framework constructed with vertices i, j, k . For example, the edge angle of the framework in Fig. 4 is given by $A(p) = [\alpha_{312}, \alpha_{413}, \alpha_{534}]$.

2.5. Strong congruency

We introduce an extension to the concept of congruency that includes the signed area or edge angle.

Lemma 1. Given two congruent Henneberg frameworks $F = (G, p)$ and $\hat{F} = (G, \hat{p})$ where $G = (\mathcal{V}, E)$, then $\chi(p) = \chi(\hat{p})$ if and only if $A(p) = A(\hat{p})$.

Proof. See Appendix A.

Definition 1. Henneberg frameworks $F = (G, p)$ and $\hat{F} = (G, \hat{p})$ are said to be *strongly congruent* if they are congruent and $\chi(p) = \chi(\hat{p})$ (or $A(p) = A(\hat{p})$).

The set of all frameworks that are strongly congruent to F will be symbolized by $\text{SCgt}(F)$. Clearly, frameworks that are congruent but not strongly congruent are *reflected* frameworks. If $\hat{F}$ is a reflected version of F , then $\chi(p) = -\chi(\hat{p})$ and $\alpha_{kij}(\hat{p}) + \alpha_{kij}(p) = 2\pi$, where $\alpha_{kij}(\hat{p})$ is any element of $A(\hat{p})$ and $\alpha_{kij}(p)$ is the corresponding element of $A(p)$. The concept of strong congruency rules out the occurrence of framework ambiguities, especially reflections.

Lemma 2. The following statements are equivalent:

- (a) Henneberg frameworks $F = (G, p)$ and $\hat{F} = (G, \hat{p})$ are strongly congruent;
- (b) Henneberg frameworks $F = (G, p)$ and $\hat{F} = (G, \hat{p})$ are equivalent, and $\chi(p) = \chi(\hat{p})$;
- (c) Henneberg frameworks $F = (G, p)$ and $\hat{F} = (G, \hat{p})$ are equivalent, and $A(p) = A(\hat{p})$.

Proof. See Appendix B.

2.6. Quartic polynomials

The following results concern polynomial equations and more specifically, quartic polynomials.

Lemma 3 (Descartes' Rule [20]). The number of positive real roots of a polynomial equation with real coefficients is either equal to the number of sign variations in the coefficients or is less than that number by a positive even integer. A root of multiplicity m is here counted as m roots.

Corollary 1. Consider the equation

$$ax^4 + bx^3 + cx^2 + dx + e = 0 \quad (9)$$

where

$$\begin{aligned} a &= -2\delta_1^2(\gamma - 2)^2 < 0 \\ b &= \delta_1(\gamma^2 - 4)\sqrt{\rho} > 0 \\ c &= -\frac{1}{2}\delta_1^4\gamma^3 + \delta_1^2\left(\frac{3}{2}\delta_1^2 + \delta_2^2 + \delta_3^2\right)\gamma^2 \\ &\quad - 4\delta_1^2(\delta_2^2 + \delta_3^2)\gamma < 0 \\ d &= \frac{1}{4}\delta_1\gamma^2(2\delta_1^2\gamma - 3\delta_1^2 - 2\delta_2^2 - 2\delta_3^2)\sqrt{\rho} > 0 \\ e &= -\frac{1}{8}\delta_1^2\gamma^3\rho < 0, \end{aligned} \quad (10)$$

$$\rho = 2\delta_1^2\delta_2^2 + 2\delta_1^2\delta_3^2 + 2\delta_2^2\delta_3^2 - \delta_1^4 - \delta_2^4 - \delta_3^4, \quad (11)$$

γ is a positive constant, and $\delta_1, \delta_2, \delta_3$ are the lengths of the edges of a triangle. If

$$\gamma > \frac{3}{2} + \frac{\delta_2^2 + \delta_3^2}{\delta_1^2} + \eta \quad (12)$$

where

$$\eta = \begin{cases} 0, & \text{if } \frac{\delta_2^2 + \delta_3^2}{\delta_1^2} < \frac{9}{2} \\ \sqrt{\left(\frac{\delta_2^2 + \delta_3^2}{\delta_1^2} - \frac{9}{2}\right)\left(\frac{\delta_2^2 + \delta_3^2}{\delta_1^2} - \frac{1}{2}\right)}, & \text{if } \frac{\delta_2^2 + \delta_3^2}{\delta_1^2} \geq \frac{9}{2}, \end{cases} \quad (13)$$

then the roots of (9) are either complex numbers or positive real numbers.

Proof. See Appendix C.

2.7. Stability results

Lemma 4 ([21]). Consider the system $\dot{x} = f(x, u)$ where x is the state, u is the control input, and $f(x, u)$ is locally Lipschitz in (x, u) in some neighborhood of $(x = 0, u = 0)$. Then, the system is locally input-to-state stable if and only if the unforced system $\dot{x} = f(x, 0)$ has a locally asymptotically stable equilibrium point at the origin.

Lemma 5 ([22]). Consider the interconnected system

$$\begin{aligned} \Sigma_1: \quad \dot{x} &= f(x, y) \\ \Sigma_2: \quad \dot{y} &= g(y). \end{aligned} \quad (14)$$

If subsystem Σ_1 with input y is locally input-to-state stable and $y = 0$ is a locally asymptotically stable equilibrium point of subsystem Σ_2 , then $[x, y] = 0$ is a locally asymptotically stable equilibrium point of the interconnected system.

The stability analysis of this paper will employ multiple Lyapunov function candidates associated with different subsystems or different regions of the state space. Since the individual functions may not decrease for all time, we will construct a switching control scheme that pieces together the multiple functions such that the overall energy of the system always decreases with time. The idea is exemplified in Fig. 5 for the case of two functions. For this reason, the following result related to the stability of switched systems is presented.

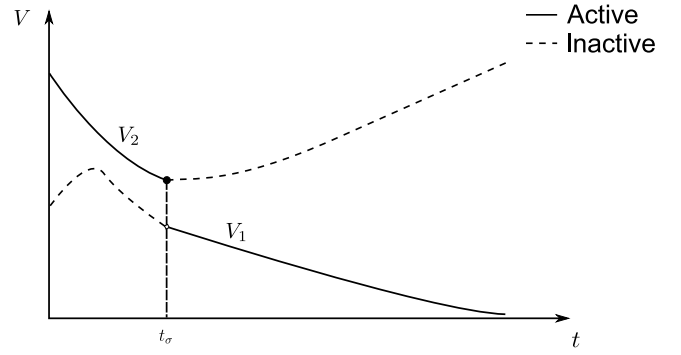


Fig. 5. Example of multiple Lyapunov function candidates versus time with a single switching time at t_σ . V_2 (subsystem 2) is active before t_σ , and V_1 (subsystem 1) is active after t_σ .

Lemma 6 ([13]).¹ Consider the family of systems

$$\dot{x} = f_q(x), \quad q \in \mathcal{Q} \quad (15)$$

where $\mathcal{Q}$ is some finite index set and each $f_q : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is locally Lipschitz and takes the triangular form

$$f_q(x) = \begin{pmatrix} f_{q1}(x_1) \\ f_{q2}(x_1, x_2) \\ \vdots \\ f_{qn}(x_1, x_2, \dots, x_n) \end{pmatrix}. \quad (16)$$

If the family of systems (15) gives rise to the switched system

$$\dot{x} = f_{\sigma(t)}(x) \quad (17)$$

where $\sigma : [0, \infty) \rightarrow \mathcal{Q}$ is the switching signal, then (17) is locally asymptotically stable at the origin if (a) (15) is locally asymptotically stable at the origin for each $q \in \mathcal{Q}$ and (b) the system

$$\dot{x}_i = f_{qi}(x_1, \dots, x_{i-1}, x_i) \quad (18)$$

is locally input-to-state stable with respect to input $u_i := [x_1, \dots, x_{i-1}]$ for each $i = 2, \dots, n$ and each $q \in \mathcal{Q}$.

3. Problem statement

Consider a group of $N \geq 3$ agents with single-integrator kinematics:

$$\dot{p}_i = u_i, \quad i = 1, \dots, N \quad (19)$$

where $p_i \in \mathbb{R}^2$ denotes the position of the i th agent relative to a global coordinate frame and $u_i \in \mathbb{R}^2$ is the control input.

The agents' desired formation is represented by the directed framework $F^* = (G, p^*)$ where $G = (\mathcal{V}, E)$, $|E| = a$, $p^* = [p_1^*, \dots, p_N^*]$, and $p_i^* \in \mathbb{R}^2$ is the desired position of agent i . The constant desired distance between agents i and j is given by

$$d_{ji} = \|p_j^* - p_i^*\| > 0, \quad i, j \in \mathcal{V}. \quad (20)$$

Note that p^* is not explicitly needed by the control since, in practice, the desired formation is specified by the desired inter-agent distances. We construct F^* with the following properties:

1. $\text{out}(1) = 0$, $\text{out}(2) = 1$, and $\text{out}(i) = 2, \forall i \geq 3$.
2. If a directed edge connects agents i and j where $i < j$, its direction should be $j \rightarrow i$.

¹ This lemma is the local version of Theorem 2.14 in [13] and is based on the discussions on page 45 of [13].

3. If $(k, i), (k, j) \in E$, then $(j, i) \in E$ (for distance + angle scheme only).

These properties suggest that F^* is a leader-first-follower, minimally persistent framework [12] with agent 1 being the leader, agent 2 the first follower, and agents i for $i \geq 3$ the ordinary followers.

We represent the actual formation of agents by $F(t) = (G, p(t))$ where $p = [p_1, \dots, p_N]$. Notice that F and F^* have the same graph, which we assume is fixed for all time. When referring to $F(t)$, the meaning of $(j, i) \in E$ is that agent j can measure its relative position to agent i , $q_j - q_i$, but not vice versa.

The problem that we set out to solve is to design u_i such that

$$F(t) \rightarrow \text{SCgt}(F^*) \text{ as } t \rightarrow \infty. \quad (21)$$

From Definition 1, (21) is equivalent to

$$\|p_j(t) - p_i(t)\| \rightarrow d_{ji} \text{ as } t \rightarrow \infty, \quad i, j \in \mathcal{V} \quad (22)$$

and either

$$\chi(p(t)) \rightarrow \chi(p^*) \text{ as } t \rightarrow \infty \quad (23a)$$

or

$$A(p(t)) \rightarrow A(p^*) \text{ as } t \rightarrow \infty. \quad (23b)$$

4. Distance + area scheme

In this section, we introduce a switching control strategy for the formation problem defined in terms of (22) and (23a).

4.1. Error variables

The control objective, when defined by distance and area constraints, will be quantified by the following two error variables. First, if the relative position of two agents is defined as $\tilde{p}_{ji} = p_j - p_i$, the distance error is given by [6]

$$z_{ji} = \|\tilde{p}_{ji}\|^2 - d_{ji}^2. \quad (24)$$

The stacked vector of all distances errors is defined as $z = [z_{21}, \dots, z_{ji}, \dots]$, $\forall (j, i) \in E$. The area error is defined as [8]

$$\tilde{S}_{ijk} = S_{ijk} - S_{ijk}^*, \quad (k, i), (k, j) \in E \quad (25)$$

where $S_{ijk} = S(p)$ with $p = [p_i, p_j, p_k]$ and $S_{ijk}^* = S(p^*)$ with $p^* = [p_i^*, p_j^*, p_k^*]$. The stacked vector of all area errors is given by $\tilde{S} = [\tilde{S}_{123}, \dots, \tilde{S}_{ijk}, \dots]$, $\forall (k, i), (k, j) \in E$.

Since F^* is typically specified in terms of the desired inter-agent distances, a useful formula for calculating S_{ijk}^* is given by [23]

$$S_{ijk}^* = \pm \sqrt{d_{ijk} (d_{ijk} - d_{ji}) (d_{ijk} - d_{ki}) (d_{ijk} - d_{kj})} \quad (26)$$

where

$$d_{ijk} = \frac{d_{ji} + d_{ki} + d_{kj}}{2}. \quad (27)$$

If the order of agents i, j, k is counterclockwise (resp., clockwise), then (26) takes the positive (resp., negative) sign.

Remark 1. Note that imposing constraints on both the signed area and edge lengths of a triangle is not redundant since two differently-shaped triangles can have the same signed area.

4.2. Control law

The control law will be dependent on the choice of potential functions associated with the error variables. To this end, we consider the potential functions

$$V_2 = \frac{\mu_2}{4} z_{21}^2 \quad (28)$$

$$V_k = \frac{\mu_k}{4} (z_{ki}^2 + z_{kj}^2) + v_k \tilde{S}_{ijk}^2 \quad (29)$$

for $k = 3, \dots, N$, $i < k, j < k, i \neq j$, and $(k, i), (k, j) \in E$, where $\mu_k > 0$, $k = 2, \dots, N$ and

$$v_k = \begin{cases} 0, & \text{if } S_{ijk} S_{ijk}^* > 0 \\ \bar{v}_k > 0, & \text{if } S_{ijk} S_{ijk}^* \leq 0 \end{cases} \quad (30)$$

for $k = 3, \dots, N$.

Based on (29) and their time derivatives, we propose the switching controller

$$u_1 = 0 \quad (31a)$$

$$u_2 = -\mu_2 z_{21} \tilde{p}_{21} \quad (31b)$$

$$u_k = -\mu_k z_{ki} \tilde{p}_{ki} - \mu_k z_{kj} \tilde{p}_{kj} - v_k \tilde{S}_{ijk} J^T (\tilde{p}_{ki} - \tilde{p}_{kj}) \quad (31c)$$

for $k = 3, \dots, N$.

Remark 2. The reasoning behind the switching strategy in (30) is as follows. If $S_{ijk} S_{ijk}^* > 0$, then both S_{ijk} and S_{ijk}^* have the same sign and therefore a distance-only controller (first two terms in (31c)) is enough to ensure convergence to the desired formation. If $S_{ijk} S_{ijk}^* \leq 0$, then S_{ijk} and S_{ijk}^* have opposite signs or agent k is collinear with agents i and j . In this case, an area error term is added to the control law (third term in (31c)) to force $S_{ijk} S_{ijk}^*$ to become positive, thereby preventing the convergence to an ambiguous formation.

Our result for this section is summarized in the following theorem.

Theorem 1. Let $\mathcal{D}_1 := \{[z, \tilde{S}] \in \mathbb{R}^{3N-5} \mid z_{21} > -d_{21}^2\}$. For any $[z(0), \tilde{S}(0)] \in \mathcal{D}_1$, the control (31) with

$$\frac{\bar{v}_k}{\mu_k} > \frac{3}{2} + \frac{d_{ki}^2 + d_{kj}^2}{d_{ji}^2} + \eta \quad (32)$$

where

$$\eta = \begin{cases} 0, & \text{if } \frac{d_{ki}^2 + d_{kj}^2}{d_{ji}^2} < \frac{9}{2} \\ \sqrt{\left(\frac{d_{ki}^2 + d_{kj}^2}{d_{ji}^2} - \frac{9}{2}\right) \left(\frac{d_{ki}^2 + d_{kj}^2}{d_{ji}^2} - \frac{1}{2}\right)}, & \text{if } \frac{d_{ki}^2 + d_{kj}^2}{d_{ji}^2} \geq \frac{9}{2} \end{cases} \quad (33)$$

renders $[z, \tilde{S}] = 0$ asymptotically stable and ensures $F(t) \rightarrow \text{SCgt}(F^*)$ as $t \rightarrow \infty$.

Remark 3. From (24), it is clear that $z_{21} \in [-d_{21}^2, \infty)$ where $z_{21} = -d_{21}^2$ corresponds to agents 1 and 2 being collocated. Thus, the above set $\mathcal{D}_1$ simply excludes the situation where $\tilde{p}_{21} = 0$.

Remark 4. The geometric meaning of (32) is that a larger control gain ratio (i.e., more weight on the area gain $\bar{v}_k$ than on the distance gain μ_k) is required as the triangle Δ_{ijk} becomes "taller" (i.e., the altitude of vertex k relative to edge (j, i) is larger). This is because taller triangles need a larger contribution from the

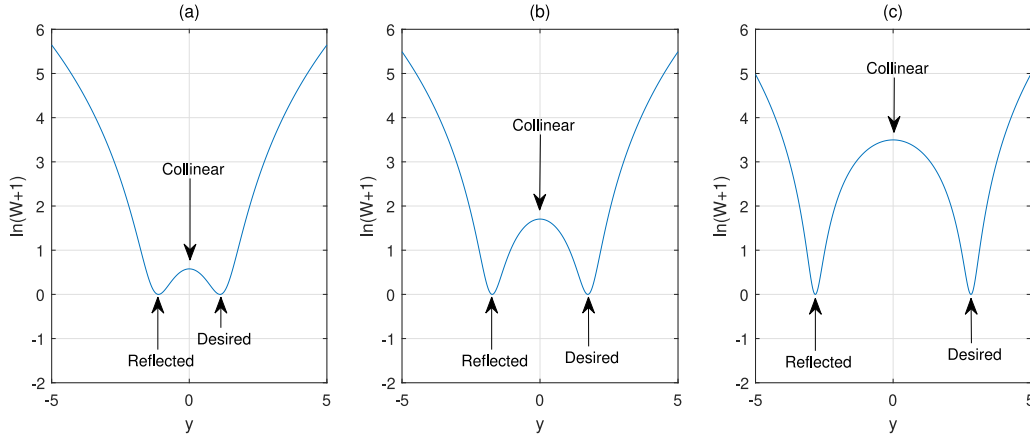


Fig. 6. Distance potential function for triangles of increasing altitude ($h_a < h_b < h_c$ where h is the altitude of vertex k).

area control term to escape the ambiguous formation. To see this, consider a triangular formation where $p_i = [-1, 0]$, $p_j = [1, 0]$, $p_k = [x, y]$, and $d_{ji} = 2$ with three different cases for the location of vertex k : a) $d_{ki} = d_{kj} = 0.75d_{ji}$, b) $d_{ki} = d_{kj} = d_{ji}$, and c) $d_{ki} = d_{kj} = 1.5d_{ji}$. Let $W := \frac{1}{4}(z_{ki}^2 + z_{kj}^2)$ be the potential function with only the distance error terms in (29). Fig. 6 shows a plot of $\ln(W+1)$ versus p_k sliced at $x = 0$ for the three cases.² We can see that the difference between the potential function value corresponding to the point where the vertices are collinear and at the local minima (desired and reflected equilibrium points) increases as the triangle grows taller. Therefore, if $S_{ijk}^* \leq 0$ (i.e., agent k is closer to the reflected equilibrium point), it will need more energy to cross the collinear position and reach the desired equilibrium point.

4.3. Proof of Theorem 1

In the following proof, we will assume $S_{ijk}^* > 0$ for simplicity of analysis so that the switching conditions in (30) become $S_{ijk} > 0$ and $S_{ijk} \leq 0$, respectively. The case where $S_{ijk}^* < 0$ can be handled with a similar analysis.

The open-loop dynamics for (24) and (25) are given by

$$\dot{z}_{ji} = 2\tilde{p}_{ji}^T(u_j - u_i) \quad (34)$$

and

$$\dot{\tilde{S}}_{ijk} = \frac{1}{2}[(u_k - u_i)^T J\tilde{p}_{kj} + \tilde{p}_{ki}^T(u_k - u_j)], \quad (35)$$

where (19) was used. Therefore, the time derivative of (29) becomes

$$\dot{V}_2 = \mu_2 z_{21} \tilde{p}_{21}^T(u_2 - u_1) \quad (36a)$$

$$\begin{aligned} \dot{V}_k = & \mu_k [z_{ki} \tilde{p}_{ki}^T(u_k - u_i) + z_{kj} \tilde{p}_{kj}^T(u_k - u_j)] \\ & + v_k \tilde{S}_{ijk} [(u_k - u_i)^T J\tilde{p}_{kj} + \tilde{p}_{ki}^T(u_k - u_j)] \end{aligned} \quad (36b)$$

for $2 < k \leq N$, $i < k$, $j < k$, $i \neq j$, and $(k, i), (k, j) \in E$.

To simplify the proof procedure, we will break up the proof into steps as we grow the number of agents in the formation analysis.

4.3.1. Step 1

First, consider the subsystem composed of agents 1 and 2 only. Substituting (31a) and (31b) into (34) yields

$$\dot{z}_{21} = -2\alpha_2 z_{21} \|\tilde{p}_{21}\|^2 = -2\mu_2 z_{21} (z_{21} + d_{21}^2) \quad (37)$$

where (24) was used. The solution to the above nonlinear ODE is

$$z_{21}(t) = \frac{d_{21}^2 z_{21}(0)}{d_{21}^2 \exp(2d_{21}^2 \mu_2 t) (z_{21}(0) + d_{21}^2) - z_{21}(0)}. \quad (38)$$

If $z_{21}(0) > -d_{21}^2$, we can show from (38) that $z_{21}(t) > -d_{21}^2 \forall t > 0$ as follows:

$$\begin{aligned} z_{21}(t) & > -d_{21}^2 \\ \Leftrightarrow \frac{z_{21}(0)}{d_{21}^2 \exp(2d_{21}^2 \mu_2 t) (z_{21}(0) + d_{21}^2) - z_{21}(0)} & > -1 \\ \Leftrightarrow z_{21}(0) > z_{21}(0) - d_{21}^2 \exp(2d_{21}^2 \mu_2 t) (z_{21}(0) + d_{21}^2) \\ \Leftrightarrow d_{21}^2 \exp(2d_{21}^2 \mu_2 t) (z_{21}(0) + d_{21}^2) & > 0 \\ \Leftrightarrow z_{21}(0) > -d_{21}^2. \end{aligned} \quad (39)$$

Now, after substituting (31a) and (31b) into (36a), we obtain

$$\dot{V}_2 = -\mu_2^2 z_{21}^2 \|\tilde{p}_{21}\|^2. \quad (40)$$

Since $z_{21}(t) > -d_{21}^2$ implies $\|\tilde{p}_{21}(t)\| > 0$, we can see that $\dot{V}_2 = 0$ only at $z_{21} = 0$. Therefore, V_2 is negative definite and $z_{21} = 0$ is asymptotically stable for $z_{21}(0) > -d_{21}^2$ (or equivalently, $p_1(0) \neq p_2(0)$).

4.3.2. Step 2

Consider that a third agent is added to the previous subsystem. We can view this new system as the triangular system

$$\dot{\xi}_3 = f_{3q}(\xi_3, \mathcal{E}_2), \quad q \in Q := \{+, -\} \quad (41a)$$

$$\dot{\mathcal{E}}_2 = g_3(\mathcal{E}_2) \quad (41b)$$

where $\xi_3 := [z_{31}, z_{32}, \tilde{S}_{123}]$ is the state of the error dynamics of agent 3 and $\mathcal{E}_2 = \xi_2 := z_{21}$. Notice that, in the light of (30), (41a) is a switched system with index set Q , where '+' (resp., '-') corresponds to switching condition $S_{ijk} > 0$ (resp., $S_{ijk} \leq 0$). Also, notice that (41b) is the same as (37).

To establish the asymptotic stability of (41) via Lemma 6, we need to check that the two lemma conditions are satisfied. To this end, when $\mathcal{E}_2 = 0$, we can see from (31b) that $u_2 = 0$. Therefore, from (36b) with $k = 3$ under the condition that $\mathcal{E}_2 = 0$, we have that

$$\dot{V}_3 = [\mu_3 (z_{31} \tilde{p}_{31} + z_{32} \tilde{p}_{32})^T + v_3 \tilde{S}_{123} (\tilde{p}_{31} - \tilde{p}_{32})^T J] u_3. \quad (42)$$

² Functions W and $\ln(W+1)$ are positively correlated; thus, this variable transformation does not change the function extrema but allows for a better visualization.

Substituting (31c) with $k = 3$ into (42) yields

$$\dot{V}_3 = - \left\| \mu_3 (z_{31}\tilde{p}_{31} + z_{32}\tilde{p}_{32}) + v_3 \tilde{S}_{123} J^T (\tilde{p}_{31} - \tilde{p}_{32}) \right\|^2. \quad (43)$$

If $\xi_3 = 0$ is the only value where (43) is zero, then $\dot{V}_3$ is negative definite. Now, note that translational and rotational motions of the triangle will not change the value of $\dot{V}_3$, since it is a function of the relative position of agents and the triangle area [24,25]. Thus, without the loss of generality, let $p_1 = [-d_{21}/2, 0]$, $p_2 = [d_{21}/2, 0]$, and $p_3 = [x, y]$ for simplicity, and recall that $S_{123}^* > 0$. Substituting this information into (2) gives $S_{123} = d_{21}y/2$; hence, the switching conditions become $y > 0$ and $y \leq 0$. Then, it follows from (43) that $\dot{V}_3 = 0$ is equivalent to

$$\begin{aligned} & \left(2x^2 + 2y^2 + \frac{d_{21}^2}{2} - d_{31}^2 - d_{32}^2 \right) x \\ & + \frac{d_{21} (2d_{21}x - d_{31}^2 - d_{32}^2)}{2} = 0 \end{aligned} \quad (44a)$$

$$\begin{aligned} & \left(2x^2 + 2y^2 + \frac{d_{21}^2}{2} - d_{31}^2 - d_{32}^2 \right) y \\ & + \frac{v_3}{\mu_3} \left(\frac{d_{21}^2}{2} y - d_{21} S_{123}^* \right) = 0 \end{aligned} \quad (44b)$$

One solution to (44) is

$$x = (d_{31}^2 - d_{32}^2) / (2d_{21}) \quad \text{and} \quad y = 2S_{123}^* / d_{21}, \quad (45)$$

which corresponds to $\xi_3 = 0$ (desired equilibrium point). We will show next that $\bar{v}_3/\mu_3$ can be selected such that this is the only solution to (44). This proof will be conducted separately for two cases: an isosceles triangle and the non-isosceles case.

Case a: Consider the triangle is such that $d_{32} = d_{31}$. From (44), we obtain

$$\left(2x^2 + 2y^2 + \frac{3d_{21}^2}{2} - 2d_{32}^2 \right) x = 0 \quad (46a)$$

$$\begin{aligned} & \left(2x^2 + 2y^2 + \frac{d_{21}^2}{2} - 2d_{32}^2 \right) y \\ & + \frac{v_3}{2\mu_3} d_{21}^2 \left(y - \frac{1}{2} \sqrt{4d_{32}^2 - d_{21}^2} \right) = 0. \end{aligned} \quad (46b)$$

When $v_3 = 0$ ($y > 0$), the only solution to (46) is

$$x = 0 \quad \text{and} \quad y = \frac{1}{2} \sqrt{4d_{32}^2 - d_{21}^2}, \quad (47)$$

which is equivalent to (45) in light of (26) and (27).

When $v_3 \neq 0$ ($y \leq 0$), we will show next that (46) does not have a solution, i.e., an (undesired) equilibrium point does not exist in this region. First, (46a) implies $x = 0$ or $x^2 + y^2 = d_{32}^2 - \frac{3}{4}d_{21}^2$. Substituting $x = 0$ into (46b) yields

$$\left(y - \frac{1}{2} \sqrt{4d_{32}^2 - d_{21}^2} \right) \left(8y^2 + 4\sqrt{4d_{32}^2 - d_{21}^2} y + \frac{2\bar{v}_3}{\mu_3} d_{21}^2 \right) = 0. \quad (48)$$

It is easy to see that when

$$\frac{\bar{v}_3}{\mu_3} > \frac{d_{32}^2 - \frac{1}{4}d_{21}^2}{d_{21}^2}, \quad (49)$$

the discriminant of $8y^2 + 4\sqrt{4d_{32}^2 - d_{21}^2} y + \frac{2\bar{v}_3}{\mu_3} d_{21}^2$ is negative. That is, inequality (49) leads to (48) not having a nonpositive solution for y .

Next, substituting

$$x^2 + y^2 = d_{32}^2 - \frac{3}{4}d_{21}^2 \quad (50)$$

into the second equation of (46b) gives

$$2y \left(\frac{\bar{v}_3}{\mu_3} - 2 \right) = \frac{\bar{v}_3}{\mu_3} \sqrt{4d_{32}^2 - d_{21}^2}. \quad (51)$$

Given (32) and the fact that $d_{21} < d_{31} + d_{32} = 2d_{32}$, we know $\frac{\bar{v}_3}{\mu_3} > \frac{3}{2} + \frac{2d_{32}^2}{d_{21}^2} > 2$. Thus, the left-hand side of (51) is non-positive while the right side is positive. This implies that (51) has no non-positive solution.

Finally, it is not difficult to show that (32) is a sufficient condition for (49). Therefore, for the isosceles triangle, the condition for $\xi_3 = 0$ to be the only value where $\dot{V}_3 = 0$ is given by (32).

Case b: Consider that the triangle is not isosceles ($d_{32} \neq d_{31}$). When $v_3 = 0$ ($y > 0$), the only solution to (44) is the desired solution (45). When $v_3 \neq 0$ ($y \leq 0$), after substituting (44a) into (44b), eliminating x , and factoring the resulting polynomial of y , we obtain

$$\left(y - \frac{2S_{123}^*}{d_{21}} \right) (c_4 y^4 + c_3 y^3 + c_2 y^2 + c_1 y + c_0) = 0 \quad (52)$$

where

$$c_4 = -2d_{21}^2 \left(\frac{\bar{v}_3}{\mu_3} - 2 \right)^2 \quad (53a)$$

$$\begin{aligned} c_3 = & d_{21} \left[\left(\frac{\bar{v}_3}{\mu_3} \right)^2 - 4 \right] \\ & \times \sqrt{2d_{21}^2 d_{32}^2 - d_{21}^4 + 2d_{21}^2 d_{31}^2 - d_{32}^4 + 2d_{32}^2 d_{31}^2 - d_{31}^4} \end{aligned} \quad (53b)$$

$$\begin{aligned} c_2 = & -\frac{1}{2} d_{21}^4 \left(\frac{\bar{v}_3}{\mu_3} \right)^3 + d_{21}^2 \left(\frac{3}{2} d_{21}^2 + d_{32}^2 + d_{31}^2 \right) \left(\frac{\bar{v}_3}{\mu_3} \right)^2 \\ & \times -4d_{21}^2 (d_{32}^2 + d_{31}^2) \frac{\bar{v}_3}{\mu_3} \end{aligned} \quad (53c)$$

$$\begin{aligned} c_1 = & \frac{1}{4} d_{21} \left(\frac{\bar{v}_3}{\mu_3} \right)^2 \left(2d_{21}^2 \frac{\bar{v}_3}{\mu_3} - 3d_{21}^2 - 2d_{32}^2 - 2d_{31}^2 \right) \times \\ & \sqrt{2d_{21}^2 d_{32}^2 - d_{21}^4 + 2d_{21}^2 d_{31}^2 - d_{32}^4 + 2d_{32}^2 d_{31}^2 - d_{31}^4} \end{aligned} \quad (53d)$$

$$\begin{aligned} c_0 = & -\frac{1}{8} d_{21}^2 \left(\frac{\bar{v}_3}{\mu_3} \right)^3 \\ & \times (2d_{21}^2 d_{32}^2 - d_{21}^4 + 2d_{21}^2 d_{31}^2 - d_{32}^4 + 2d_{32}^2 d_{31}^2 - d_{31}^4). \end{aligned} \quad (53e)$$

Note that the quartic polynomial in (52) is similar to (9), so by Corollary 1 and (32) the quartic polynomial has no non-positive real roots. This means that no equilibrium point exists when $\bar{v}_3 \neq 0$. Therefore, for the non-isosceles triangle, (32) ensures that $\xi_3 = 0$ is the only value at which $\dot{V}_3 = 0$.

Now that we have established $\dot{V}_3$ is negative definite with respect to ξ_3 when $\mathcal{E}_2 = 0$, we know that the unforced systems $\dot{\xi}_3 = f_{3+}(\xi_3, 0)$ and $\dot{\xi}_3 = f_{3-}(\xi_3, 0)$ are asymptotically stable at the origin. Then, by Lemma 4, we know (41a) is input-to-state stable with respect to $\mathcal{E}_2$ for each $q \in \mathcal{Q}$. We can now invoke Lemma 5 to claim that (41) is asymptotically stable at $[\xi_3, \mathcal{E}_2] = 0$ for each $q \in \mathcal{Q}$. Since the conditions of Lemma 6 are satisfied, it follows that the origin of switched system (41) is asymptotically stable.

4.3.3. Step $k-1$

The process of adding a vertex k with two outgoing edges to any two distinct vertices i and j of the previous graph can be followed one step at a time, resulting at each step in the switched triangular system

$$\dot{\xi}_k = f_{kq}(\xi_k, \mathcal{E}_{k-1}), \quad q \in \mathcal{Q} \quad (54a)$$

$$\dot{\mathcal{E}}_{k-1} = g_{k\bar{q}}(\mathcal{E}_{k-1}), \quad \bar{q} \in \mathcal{Q} \quad (54b)$$

where $\xi_k := [z_{ki}, z_{kj}, \tilde{s}_{ijk}]$, $(k, i), (k, j) \in E$ is the state of error dynamics of the k th agent and $\mathcal{E}_{k-1} := [\xi_2, \dots, \xi_{k-1}]$.

Note that the asymptotic stability of $\mathcal{E}_{k-1} = 0$ for (54b) was already established in Step $k-2$. Therefore, we only need to check the input-to-state stability of (54a) with respect to input $\mathcal{E}_{k-1}$. To this end, when $\mathcal{E}_{k-1} = 0$, (36b) becomes

$$\dot{V}_k = \left[\mu_k (z_{ki}\tilde{p}_{ki} + z_{kj}\tilde{p}_{kj})^\top + \nu_k \tilde{s}_{ijk} (\tilde{p}_{ki} - \tilde{p}_{kj})^\top \right] u_k. \quad (55)$$

Now, substituting (31c) into (55) gives

$$\dot{V}_k = - \left\| \mu_k (z_{ki}\tilde{p}_{ki} + z_{kj}\tilde{p}_{kj}) + \nu_k \tilde{s}_{ijk} J^\top (\tilde{p}_{ki} - \tilde{p}_{kj}) \right\|^2. \quad (56)$$

Similar to Step 2, we can show that if the gain ratio $\tilde{\nu}_k/\mu_k$ is selected according to (32), then (56) is negative definite with respect to ξ_k . As a result, unforced systems $\dot{\xi}_k = f_{k+}(\xi_k, 0)$ and $\dot{\xi}_k = f_{k-}(\xi_k, 0)$ are asymptotically stable at $\xi_k = 0$ and (54a) is input-to-state stable with respect to $\mathcal{E}_{k-1}$ for each $q \in \mathcal{Q}$ by Lemma 4. We now know $[\mathcal{E}_{k-1}, \xi_k] = 0$ in (54) is asymptotically stable for each $q \in \mathcal{Q}$ from Lemma 5. Finally, we can use Lemma 6 to show that switched system (54) is asymptotically stable at the origin.

Repeating this process until $k = N$ leads to the conclusion that $[\xi_2, \dots, \xi_N] = 0$ is asymptotically stable, which implies $z(t) \rightarrow 0$ and $\chi(p(t)) \rightarrow \chi(p^*)$ as $t \rightarrow \infty$. Given that F^* and $F(t)$ have the same edge set and F^* is minimally persistent by design, then we have that $F(t) \rightarrow \text{SCgt}(F^*)$ as $t \rightarrow \infty$ from Lemma 2.

5. Distance + angle scheme

Here, the switching control strategy is recast in order to satisfy (22) and (23b).

5.1. Error variables

In the case where the control objective is defined by distance and angle constraints, we use the following modified distance error (see Remark 5)

$$e_{ji} = \|\tilde{p}_{ji}\| - d_{ij}, \quad (57)$$

where the vector of all distances errors is given by $e = [e_{21}, \dots, e_{ji}, \dots]$, $\forall (j, i) \in E$. The edge angle error is defined as

$$\tilde{\alpha}_{kij} = \alpha_{kij} - \alpha_{kij}^* \quad (58)$$

with the following revised definition of the edge angle

$$\alpha_{kij}(p) = \begin{cases} 2\pi - \text{mod}\{-(\alpha_{kj} - \alpha_{ki}), 2\pi\} \in (0, 2\pi], & \text{if } 0 < \alpha_{kij}^* < \pi \\ \text{mod}\{(\alpha_{kj} - \alpha_{ki}), 2\pi\} \in [0, 2\pi), & \text{if } \pi < \alpha_{kij}^* < 2\pi, \end{cases} \quad (59)$$

where α_{ki} and α_{kj} were defined in (7). Note from (59) that when $\alpha_{kij} = 2m\pi$, $m \in \{0, 1\}$, we have $\alpha_{kij} = 2\pi$ in the first case and $\alpha_{kij} = 0$ in the second case. The reason for defining the edge angle as in (59) will be explained later in Remark 6.

The vector of all edge angle errors is given by $\tilde{\alpha} = [\tilde{\alpha}_{312}, \dots, \tilde{\alpha}_{kij}, \dots]$, $\forall (k, i), (k, j) \in E$. Since F^* is typically specified in terms of the desired inter-agent distances, α_{kij}^* can be calculated using the law of cosines as follows

$$\alpha_{kij}^* = \text{mod}\left\{\pm \arccos\left(\frac{d_{ki}^2 + d_{kj}^2 - d_{ji}^2}{2d_{ki}d_{kj}}\right), 2\pi\right\}. \quad (60)$$

Note that $\alpha_{kij}^* \in (0, 2\pi)$, and if the order of agents k, i, j is counterclockwise (resp., clockwise), then (60) takes the positive (resp., negative) sign.

Remark 5. Since the definition of the edge angle requires the length of each edge to be nonzero for all time, we will need to prove that the control law ensures $\|p_{ji}(t)\| > 0$, $\forall (j, i) \in E$ and $\forall t > 0$. This proof is facilitated by using (57) instead of (24) as the measure of the distance error for the distance + angle scheme.

5.2. Control law

For the distance + angle scheme, we introduce the potential functions

$$V_2 = \mu_2 \phi_{21} \quad (61)$$

$$V_k = \mu_k (\phi_{ki} + \phi_{kj}) + \frac{\nu_k}{2} \tilde{\alpha}_{kij}^2 \quad (62)$$

for $k = 3, \dots, N$, $i < k, j < k$, $i \neq j$, and $(k, i), (k, j) \in E$, where $\mu_k > 0$, $k = 2, \dots, N$,

$$\phi_{ki} = \frac{e_{ki}(e_{ki} + 2d_{ki})}{2d_{ki}^2} - \ln \frac{(e_{ki} + d_{ki})}{d_{ki}}, \quad (63)$$

$$\nu_k = \begin{cases} 0, & \text{if } (\alpha_{kij} - \pi)(\alpha_{kij}^* - \pi) > 0 \\ \tilde{\nu}_k > 0, & \text{if } (\alpha_{kij} - \pi)(\alpha_{kij}^* - \pi) \leq 0 \end{cases} \quad (64)$$

along with the switching control law

$$u_1 = 0 \quad (65a)$$

$$u_2 = -\mu_2 w_{21} \tilde{p}_{21} \quad (65b)$$

$$u_k = -\mu_k w_{ki} \tilde{p}_{ki} - \mu_k w_{kj} \tilde{p}_{kj} - \nu_k \tilde{\alpha}_{kij} J^\top \left(\frac{\tilde{p}_{kj}}{\|\tilde{p}_{kj}\|^2} - \frac{\tilde{p}_{ki}}{\|\tilde{p}_{ki}\|^2} \right) \quad (65c)$$

for $k = 3, \dots, N$, where

$$w_{ki} = \frac{1}{d_{ki}^2} - \frac{1}{\|\tilde{p}_{ki}\|^2}. \quad (66)$$

Remark 6. We use the sign of $(\alpha_{kij} - \pi)(\alpha_{kij}^* - \pi)$ to distinguish two regions of the state space. Specifically, if $\alpha_{kij}^* \in (0, \pi)$, we use the distance-only controller when $\alpha_{kij} \in (0, \pi)$ (i.e., $(\alpha_{kij} - \pi)(\alpha_{kij}^* - \pi) > 0$) since agent k is on the “correct side” with respect to the desired formation, and use the controller with the additional angle error term when $\alpha_{kij} \in [\pi, 2\pi]$ (i.e., $(\alpha_{kij} - \pi)(\alpha_{kij}^* - \pi) \leq 0$) since agent k is either on the “incorrect side” or collinear with agents i and j . To simplify the switching logic, α_{kij} is calculated here using the first case of (59) so that $\alpha_{kij} = 0$ is converted to 2π and included in the region $\alpha_{kij} \in [\pi, 2\pi]$. If $\alpha_{kij}^* \in (\pi, 2\pi)$, we use the distance-only controller when $\alpha_{kij} \in (\pi, 2\pi)$ (i.e., $(\alpha_{kij} - \pi)(\alpha_{kij}^* - \pi) > 0$), and the distance + angle controller when $\alpha_{kij} \in [0, \pi]$ (i.e., $(\alpha_{kij} - \pi)(\alpha_{kij}^* - \pi) \leq 0$). For $\alpha_{kij}^* \in (\pi, 2\pi)$, α_{kij} is calculated by the second case in (59) so that $\alpha_{kij} = 2\pi$ is converted to 0 and included in the region $\alpha_{kij} \in [0, \pi]$. See Fig. 7 for an illustration of the switching logic.

To facilitate the control analysis, we will first present some preliminary results.

Lemma 7. Consider a system composed of agents i, j , and k , where $(k, i), (k, j), (j, i) \in E$. If $u_i = u_j = 0$, then $\tilde{p}_{ki}(t)$ and $\tilde{p}_{kj}(t)$ are bounded for all time.

Proof. After taking the derivative of (62), we obtain

$$\begin{aligned} \dot{V}_k = & \mu_k [w_{ki} \tilde{p}_{ki}^\top (u_k - u_i) + w_{kj} \tilde{p}_{kj}^\top (u_k - u_j)] \\ & + \nu_k \tilde{\alpha}_{kij} \left[\frac{\tilde{p}_{kj}^\top J}{\|\tilde{p}_{kj}\|^2} (u_k - u_j) - \frac{\tilde{p}_{ki}^\top J}{\|\tilde{p}_{ki}\|^2} (u_k - u_i) \right]. \end{aligned} \quad (67)$$

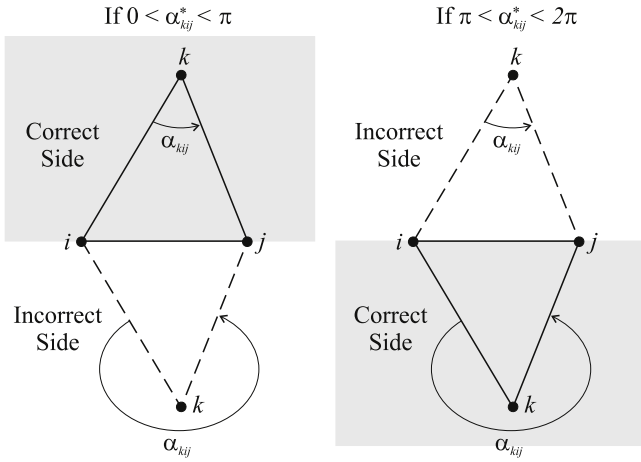


Fig. 7. Graphical representation of the switching logic.

Substituting $u_i = u_j = 0$ and (65c) into (67) gives

$$\dot{V}_k = - \left\| \mu_k (w_{ki} \tilde{p}_{ki} + w_{kj} \tilde{p}_{kj}) + v_k \tilde{\alpha}_{kij} \left(\frac{J^T \tilde{p}_{kj}}{\|\tilde{p}_{kj}\|^2} - \frac{J^T \tilde{p}_{ki}}{\|\tilde{p}_{ki}\|^2} \right) \right\|^2 \leq 0. \quad (68)$$

Since (68) is nonpositive, we know that $V_k(t)$ is bounded for all time and thus, so are $(e_{ki}(t), e_{kj}(t), \tilde{\alpha}_{kij}(t))$. From (57), we then know $(\tilde{p}_{ki}(t), \tilde{p}_{kj}(t))$ are bounded for all time. ■

Lemma 8. If $\tilde{p}_{ji}(0) \neq 0$, $\forall (j, i) \in E$ and $\tilde{p}_{ji}(t)$ for $j > 2$ is bounded for all time, then (65) ensures $\tilde{p}_{ji}(t) \neq 0$, $\forall t > 0$ and $\forall (j, i) \in E$.

Proof. First, it follows from (57) that

$$\tilde{p}_{ji} \neq 0 \Leftrightarrow \|\tilde{p}_{ji}\| > 0 \Leftrightarrow e_{ji} > -d_{ji}. \quad (69)$$

After differentiating (57), we obtain the following distance error dynamics

$$\dot{e}_{ji} = \frac{\tilde{p}_{ji}^T (u_j - u_i)}{\|\tilde{p}_{ji}\|}. \quad (70)$$

Now, consider the subsystem comprised of only agents 1 and 2. If we substitute (65a) and (65b) into (70), we obtain

$$\dot{e}_{21} = -\mu_2 w_{21} \|\tilde{p}_{21}\| = -\mu_2 \frac{e_{21}(e_{21} + 2d_{21})}{d_{21}^2(e_{21} + d_{21})} \quad (71)$$

where (57) and (66) were used. Since $e_{21} \geq -d_{21}$, we can see that $\dot{e}_{21} > 0$ if $e_{21} < 0$, $\dot{e}_{21} < 0$ if $e_{21} > 0$, and $\dot{e}_{21} = 0$ if and only if $e_{21} = 0$. Therefore, $e_{21}(t) > -d_{21}$, $\forall t > 0$ if $e_{21}(0) > -d_{21}$, and $e_{21} = 0$ is the unique asymptotically stable equilibrium point of (71). From (69), we then know that $\tilde{p}_{21}(t) \neq 0$, $\forall t > 0$. Since $\tilde{p}_{21}$ cannot be zero, we know from (66) that $w_{21}(t)$ is bounded for all time. Since (71) is asymptotically stable at the origin, we know that $e_{21}(t)$ is bounded for all time; thus, $\tilde{p}_{21}(t)$ is bounded for all time from (57). It then follows from (65b) that $u_2(t)$ is bounded for all time.

Now, consider that a third agent along with edges (3, 1) and (3, 2) is added to the previous subsystem. To show the non-collision of agents 3 and 2, we substitute (65b) and (65c) with $k = 3$ for condition $0 < \alpha_{kij} < \pi$ into (70) to obtain

$$\begin{aligned} \dot{e}_{32} &= \frac{\tilde{p}_{32}}{\|\tilde{p}_{32}\|} \left[-\mu_3 (w_{31} \tilde{p}_{31} + w_{32} \tilde{p}_{32}) - u_2 \right] \\ &= -\mu_3 w_{32} \|\tilde{p}_{32}\| - \mu_3 w_{31} \|\tilde{p}_{31}\| \cos \alpha_{312} - \|u_2\| \cos \theta_a \end{aligned}$$

$$= -\frac{1}{d_{31}^2(e_{32} + d_{32})} [\mu_3 e_{32}^2 + (2\mu_3 d_{32} + d_{32}^2 h_a) e_{32} + d_{32}^3 h_a] \quad (72)$$

where θ_a denotes the angle between vectors $\tilde{p}_{32}$ and u_2 , and

$$h_a := \mu_3 w_{31} \|\tilde{p}_{31}\| \cos \alpha_{312} + \|u_2\| \cos \theta_a. \quad (73)$$

The roots of the bracketed, quadratic term in (72) are

$$\begin{aligned} r_{a1}^* &= -d_{32} - \frac{d_{32}(h_a d_{32} + \sqrt{h_a^2 d_{32}^2 + 4\mu_3^2})}{2\mu_3} < -d_{32} \\ r_{a2}^* &= -d_{32} - \frac{d_{32}(h_a d_{32} - \sqrt{h_a^2 d_{32}^2 + 4\mu_3^2})}{2\mu_3} > -d_{32}. \end{aligned} \quad (74)$$

Notice that (73) may go to infinity if $\tilde{p}_{31} \rightarrow 0$ or $\|u_2\| \rightarrow \infty$. However, we showed above that u_2 is bounded. Now, assume that $\tilde{p}_{32}$ is approaching zero. Since we already know that $\tilde{p}_{21}(t) \neq 0$, $\forall t > 0$, then $\tilde{p}_{31}$ cannot approach zero simultaneously with $\tilde{p}_{32}$. Therefore, (73) is bounded. From (72), we know that if $e_{32}(0) > -d_{32}$, then $\dot{e}_{32}(t) > 0$, $\forall t > 0$ when $-d_{32} < e_{32}(t) < r_{a2}^*$, and $\dot{e}_{32}(t) < 0$, $\forall t > 0$ when $e_{32}(t) > r_{a2}^*$. This implies $e_{32}(t) > -d_{32}$ (i.e., $\tilde{p}_{32}(t) \neq 0$), $\forall t > 0$ if $e_{32}(0) > -d_{32}$. An analogous analysis applied to the dynamics of e_{31} leads to the conclusion that $\tilde{p}_{31}(t) \neq 0$, $\forall t > 0$. Since $0 < \|\tilde{p}_{31}\| < \infty$ and $0 < \|\tilde{p}_{32}\| < \infty$, we know w_{31} , w_{32} , and $\tilde{\alpha}_{312}$ are bounded from (60) and (66). Therefore, u_3 is bounded from (65c).

Now, after substituting (65b) and (65c) with $k = 3$ for condition $\pi \leq \alpha_{kij} \leq 2\pi$ into (70), we obtain

$$\begin{aligned} \dot{e}_{32} &= \frac{\tilde{p}_{32}}{\|\tilde{p}_{32}\|} \left[-\mu_3 (w_{31} \tilde{p}_{31} + w_{32} \tilde{p}_{32}) - v_3 \tilde{\alpha}_{312} J^T \right. \\ &\quad \left. \left(\frac{\tilde{p}_{32}}{\|\tilde{p}_{32}\|^2} - \frac{\tilde{p}_{31}}{\|\tilde{p}_{31}\|^2} \right) - u_2 \right] \\ &= -\mu_3 w_{32} \|\tilde{p}_{32}\| - \mu_3 w_{31} \|\tilde{p}_{31}\| \cos \alpha_{312} \\ &\quad + v_3 \tilde{\alpha}_{312} \frac{\sin \alpha_{312}}{\|\tilde{p}_{31}\|} - \|u_2\| \cos \theta_b \\ &= -\frac{1}{d_{31}^2(e_{32} + d_{32})} [\mu_3 e_{32}^2 + (2\mu_3 d_{32} + d_{32}^2 h_b) e_{32} + d_{32}^3 h_b] \end{aligned} \quad (75)$$

where θ_b denotes the angle between vectors $\tilde{p}_{32}$ and u_2 , and

$$h_b := \mu_3 w_{31} \|\tilde{p}_{31}\| \cos \alpha_{312} - v_3 \tilde{\alpha}_{312} \frac{\sin \alpha_{312}}{\|\tilde{p}_{31}\|} + \|u_2\| \cos \theta_b. \quad (76)$$

The roots of the bracketed, quadratic term in (75) are

$$\begin{aligned} r_{b1}^* &= -d_{32} - \frac{d_{32}(h_b d_{32} + \sqrt{h_b^2 d_{32}^2 + 4\mu_3^2})}{2\mu_3} < -d_{32} \\ r_{b2}^* &= -d_{32} - \frac{d_{32}(h_b d_{32} - \sqrt{h_b^2 d_{32}^2 + 4\mu_3^2})}{2\mu_3} > -d_{32}. \end{aligned} \quad (77)$$

Similar to the case with condition $0 < \alpha_{kij} < \pi$, we can show that the $e_{32}(t) > -d_{32}$ (i.e., $\tilde{p}_{32}(t) \neq 0$), $\forall t > 0$ if $e_{32}(0) > -d_{32}$. An analogous analysis applied to the dynamics of e_{31} leads to the conclusion that $\tilde{p}_{31}(t) \neq 0$, $\forall t > 0$. Since $0 < \|\tilde{p}_{31}\| < \infty$ and $0 < \|\tilde{p}_{32}\| < \infty$, we know w_{31} , w_{32} , and $\tilde{\alpha}_{312}$ are bounded from (60) and (66). Therefore, u_3 is bounded from (65c).

Next, consider that we recursively grow the graph by adding an agent and two edges to the graph of the previous iteration. Therefore, we have the general case of agent pairs (k, i) and (k, j) where $k \geq 3, j \geq 2, i \geq 1$, and $k > j > i$. To show the non-collision of agents k and i (j), we can analyze the dynamics of e_{ki} (e_{kj}) as

was done above for e_{32} . Recall that we required $\tilde{p}_{21} \neq 0$ to avoid the situation where $\tilde{p}_{31}$ and $\tilde{p}_{32}$ approach zero simultaneously. Similarly, we now require that $\tilde{p}_{ji} \neq 0$ so $\tilde{p}_{ki}$ and $\tilde{p}_{kj}$ do not approach zero simultaneously. Given Property 3 (see Section 3), there is an edge (j, i) and thus we can show that $0 < \|\tilde{p}_{ji}\| < \infty$ in the previous step of the recursive procedure. This recursive analysis process is repeated until $k = N$. ■

Lemma 9. *The following equation*

$$\mu_k(w_{ki}\tilde{p}_{ki} + w_{kj}\tilde{p}_{kj}) + v_k\tilde{\alpha}_{kij} \left(\frac{J^T\tilde{p}_{kj}}{\|\tilde{p}_{kj}\|^2} - \frac{J^T\tilde{p}_{ki}}{\|\tilde{p}_{ki}\|^2} \right) = 0 \quad (78)$$

has a unique solution if $\|\tilde{p}_{ji}\| = d_{ji}$, $\|\tilde{p}_{ki}(t)\| \neq 0$, $\|\tilde{p}_{kj}(t)\| \neq 0$, and

$$\frac{\tilde{v}_k}{\mu_k} \geq \frac{1 + \cos \alpha_{kij}^*}{(2\pi - 2\alpha_{kij}^*) \sin \alpha_{kij}^*}. \quad (79)$$

Proof. See Appendix D.

The following theorem states the main result of this section.

Theorem 2. *Let $\mathcal{D}_2 := \{[e, \tilde{\alpha}] \in \mathbb{R}^{2N-3} \times (-2\pi, 2\pi)^{N-2} \mid e_{ji} > -d_{ji}, \forall (j, i) \in E\}$. For any $[e(0), \tilde{\alpha}(0)] \in \mathcal{D}_2$, the control (65) with (79) ensures the asymptotic stability of $[e, \tilde{\alpha}] = 0$ and $F(t) \rightarrow \text{SCgt}(F^*)$ as $t \rightarrow \infty$.*

Remark 7. Note that the set $\mathcal{D}_2$ excludes the situation where $\tilde{p}_{ji} = 0$, $\forall (j, i) \in E$. The gain condition in (79) has a similar geometric interpretation as (32). That is, larger control gain ratios are required for taller triangles $\triangle jik$. Indeed, the right-hand side of (79) approaches 0.25 as $\alpha_{kij}^* \rightarrow \pi$ (flat triangle) and approaches positive infinity as $\alpha_{kij}^* \rightarrow 0, 2\pi$ (tall triangle).

5.3. Proof of Theorem 2

To simplify the presentation, we will assume $0 < \alpha_{kij}^* < \pi$ so that the switching conditions become $0 < \alpha_{kij} < \pi$ and $\pi \leq \alpha_{kij} \leq 2\pi$, respectively. The case where $\pi < \alpha_{kij}^* < 2\pi$ can be handled with an analysis similar to the one that follows.³

We begin by computing the open-loop dynamics for (57) and (58):

$$\dot{e}_{ji} = \frac{\tilde{p}_{ji}^T(u_j - u_i)}{\|\tilde{p}_{ji}\|} \quad (80)$$

and

$$\dot{\alpha}_{kij} = \frac{\tilde{p}_{kj}^T J}{\|\tilde{p}_{kj}\|^2}(u_k - u_j) - \frac{\tilde{p}_{ki}^T J}{\|\tilde{p}_{ki}\|^2}(u_k - u_i). \quad (81)$$

Differentiating (62) and substituting (80) and (81) yields

$$\dot{V}_2 = \mu_2 w_{21} \tilde{p}_{21}^T(u_2 - u_1) \quad (82a)$$

$$\begin{aligned} \dot{V}_k = & \mu_k [w_{ki} \tilde{p}_{ki}^T(u_k - u_i) + w_{kj} \tilde{p}_{kj}^T(u_k - u_j)] \\ & + v_k \tilde{\alpha}_{kij} \left[\frac{\tilde{p}_{kj}^T J}{\|\tilde{p}_{kj}\|^2}(u_k - u_j) - \frac{\tilde{p}_{ki}^T J}{\|\tilde{p}_{ki}\|^2}(u_k - u_i) \right]. \end{aligned} \quad (82b)$$

5.3.1. Step 1 (agents 1 and 2)

Recall that we already showed during the proof of Lemma 8 that $e_{21} = 0$ is asymptotically stable and $0 < \|\tilde{p}_{21}(t)\| < \infty$, $\forall t > 0$ when $\tilde{p}_{21}(0) \neq 0$.

³ Note that $\alpha_{kij}^* = 0, \pi$, or 2π corresponds to a desired framework where the agents are collinear, which is not minimally persistent. Thus, these cases are excluded from our analysis.

5.3.2. Step 2 (agents 1, 2, and 3)

Consider the switched triangular system

$$\dot{\xi}_3 = f_{3q}(\xi_3, \mathcal{E}_2), \quad q \in \mathcal{Q} \quad (83a)$$

$$\dot{\mathcal{E}}_2 = g_3(\mathcal{E}_2) \quad (83b)$$

where $\xi_3 := [e_{31}, e_{32}, \tilde{\alpha}_{312}]$ is the state of the error dynamics of agent 3, $\mathcal{E}_2 = \xi_2 := e_{21}$, (83b) is the same as (71), and the switching index set $\mathcal{Q}$ is such that ‘+’ (resp., ‘−’) corresponds to switching condition $0 < \alpha_{kij} < \pi$ (resp., $\pi \leq \alpha_{kij} \leq 2\pi$).

In order to use Lemma 6, we will show that its two conditions are met. It is not difficult to check using (57) and (66) that $w_{ji} = 0$ if and only if $e_{ji} = 0$. Thus, we can see from (65b) that $\mathcal{E}_2 = 0 \Rightarrow u_2 = 0$ since $\tilde{p}_{21}$ is bounded. From (82b) with $k = 3$ and $\mathcal{E}_2 = 0$, we then have that

$$\dot{V}_3 = \left[\mu_3(w_{31}\tilde{p}_{31}^T + w_{32}\tilde{p}_{32}^T) + v_3\tilde{\alpha}_{312} \left(\frac{\tilde{p}_{32}^T J}{\|\tilde{p}_{32}\|^2} - \frac{\tilde{p}_{31}^T J}{\|\tilde{p}_{31}\|^2} \right) \right] u_3. \quad (84)$$

After substituting (65c) for $k = 3$ in (84), we have

$$\begin{aligned} \dot{V}_3 = & - \left\| \mu_3(w_{31}\tilde{p}_{31} + w_{32}\tilde{p}_{32}) + v_3\tilde{\alpha}_{312} J^T \left(\frac{\tilde{p}_{32}}{\|\tilde{p}_{32}\|^2} - \frac{\tilde{p}_{31}}{\|\tilde{p}_{31}\|^2} \right) \right\|^2. \end{aligned} \quad (85)$$

In order for $\dot{V}_3 = 0$, we obviously need

$$\mu_3(w_{31}\tilde{p}_{31} + w_{32}\tilde{p}_{32}) + v_3\tilde{\alpha}_{312} J^T \left(\frac{\tilde{p}_{32}}{\|\tilde{p}_{32}\|^2} - \frac{\tilde{p}_{31}}{\|\tilde{p}_{31}\|^2} \right) = 0. \quad (86)$$

Since $u_1 = u_2 = 0$, we know $\tilde{p}_{31}$ and $\tilde{p}_{32}$ are bounded from Lemma 7. We then know $\tilde{p}_{31}(t) \neq 0$ and $\tilde{p}_{32}(t) \neq 0$ for all time from Lemma 8. Finally, we can use Lemma 9 to show that (86) has a unique solution at the equilibrium point $\xi_3 = 0$ and V_3 is negative definite with respect to ξ_3 .

Due to the negative definiteness of $\dot{V}_3$ when $\mathcal{E}_2 = 0$, unforced systems $\dot{\xi}_3 = f_{3+}(\xi_3, 0)$ and $\dot{\xi}_3 = f_{3-}(\xi_3, 0)$ have an asymptotically stable equilibrium at $\xi_3 = 0$. From Lemma 4, we then know (83a) is input-to-state stable with respect to $\mathcal{E}_2$ for each $q \in \mathcal{Q}$. Next, we know from Lemma 5 that (83) has an asymptotically stable equilibrium at $[\xi_3, \mathcal{E}_2] = 0$ for each $q \in \mathcal{Q}$. Finally, we can claim that the origin of switched system (83) is asymptotically stable from Lemma 6.

5.3.3. Step $k-1$ (Agents 1, ..., k)

At this step, we have an interconnected system similar to the one in (54) with $\xi_k := [e_{ki}, e_{kj}, \tilde{\alpha}_{kij}]$, $(k, i), (k, j) \in E$ as the state of error dynamics of the k th agent. Since the asymptotic stability of $\mathcal{E}_{k-1} = 0$ for (54b) was established in Step $k-2$, we will check the input-to-state stability of (54a) with respect to input $\mathcal{E}_{k-1}$. When $\mathcal{E}_{k-1} = 0$, (82b) simplifies to

$$\dot{V}_k = \left[\mu_k(w_{ki}\tilde{p}_{ki}^T + w_{kj}\tilde{p}_{kj}^T) + v_k\tilde{\alpha}_{kij} \left(\frac{\tilde{p}_{kj}^T J}{\|\tilde{p}_{kj}\|^2} - \frac{\tilde{p}_{ki}^T J}{\|\tilde{p}_{ki}\|^2} \right) \right] u_k. \quad (87)$$

After substituting (65c) in (87), we obtain

$$\dot{V}_k = - \left\| \mu_k(w_{ki}\tilde{p}_{ki} + w_{kj}\tilde{p}_{kj}) + v_k\tilde{\alpha}_{kij} \left(\frac{J^T\tilde{p}_{kj}}{\|\tilde{p}_{kj}\|^2} - \frac{J^T\tilde{p}_{ki}}{\|\tilde{p}_{ki}\|^2} \right) \right\|^2. \quad (88)$$

Similar to Step 2, we can show that if the gain ratio $\tilde{v}_k/\mu_k$ is selected according to (79), then (88) is negative definite with respect to ξ_k . This indicates that unforced systems $\dot{\xi}_k = f_{k+}(\xi_k, 0)$ and $\dot{\xi}_k = f_{k-}(\xi_k, 0)$ are asymptotically stable at $\xi_k = 0$ and

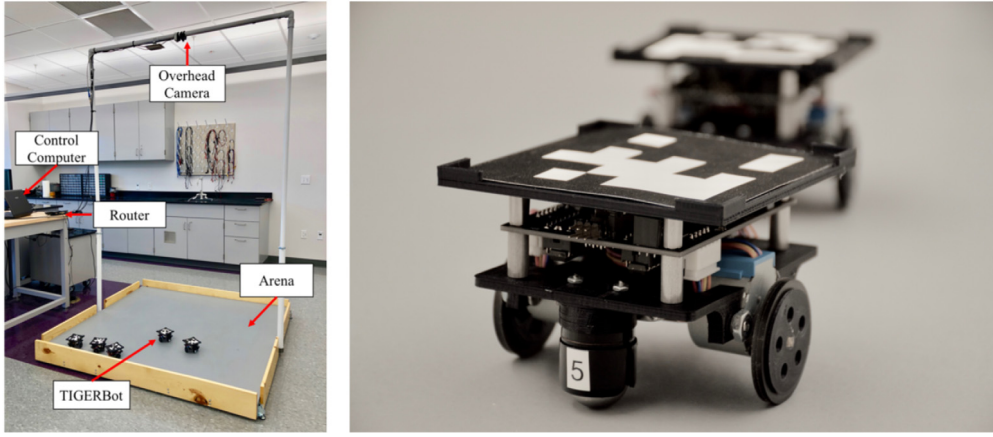


Fig. 8. TIGER Square testbed (left) and TIGERBot (right).

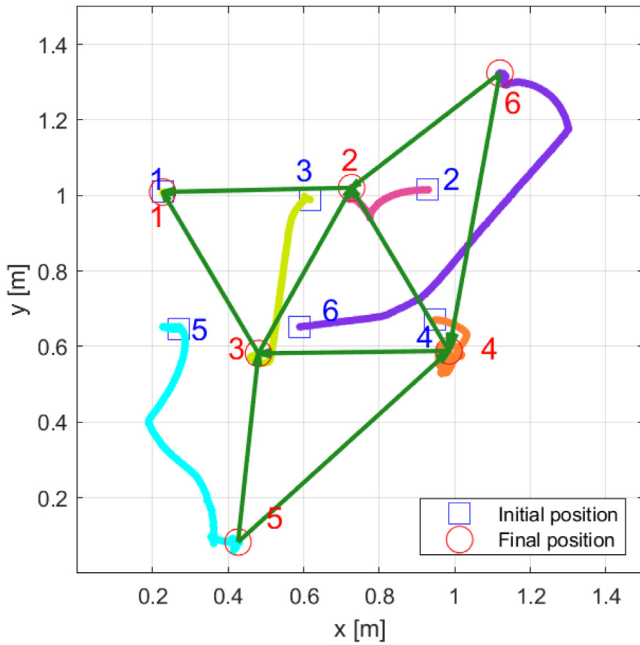


Fig. 9. Distance + area control: Agent trajectories $p_i(t)$, $i = 1, \dots, 6$ for $t = [0, 65]$ s.

(54a) is input-to-state stable for each q from Lemma 4. It then follows from Lemma 5 that $[\xi_{k-1}, \xi_k] = 0$ is asymptotically stable for each q in (54). Lastly, by Lemma 6, switched system (54) is asymptotically stable at the origin.

As this process is repeated until $k = N$, we conclude that $[\xi_2, \dots, \xi_N] = 0$ is asymptotically stable, indicating that $e(t) \rightarrow 0$ and $A(p(t)) \rightarrow A(p^*)$ as $t \rightarrow \infty$. Since F^* and $F(t)$ have the same edge set and F^* is minimally persistent, we know from Lemma 2 that $F(t) \rightarrow \text{SCgt}(F^*)$ as $t \rightarrow \infty$.

6. Experimental validation

The controllers proposed in Sections 4 and 5 were experimentally tested on the TIGER Square system [26] located at Louisiana State University. TIGER Square, shown in Fig. 8, is an experimental testbed for multi-agent systems, which uses the TIGERBot as the mobile robot platform. TIGERBot is a low-cost, custom-built, differential-drive robot with an approximate footprint of 89 mm $\times$ 89 mm. Each robot wirelessly interacts with a central computer

to receive control inputs and output data from onboard sensors. Currently, the testing environment is limited to 1.5 m $\times$ 1.5 m area, utilizes an overhead camera to localize the robots, and can host up to 10 robots.

The control objective for the formation shape experiments were for six agents to form the green triangles in Figs. 9 and 11. Nine directed edges were used to ensure the framework was minimally persistent. Thus, the edge set was

$$E = \{(2, 1), (3, 1), (3, 2), (4, 2), (4, 3), (5, 3), (5, 4), (6, 2), (6, 4)\}, \quad (89)$$

and $d_{21} = d_{31} = d_{32} = d_{42} = d_{43} = d_{53} = d_{62} = 0.5m$, $d_{54} = d_{64} = 0.75m$. The control gains were set to $\mu_i = 6$ and $\bar{v}_3 = \bar{v}_4 = 24$, $\bar{v}_5 = \bar{v}_6 = 36$ in the distance + area controller (31), where the ratio $\bar{v}_3/\mu_3 = \bar{v}_4/\mu_4 = 4$ is greater than 3.5 calculated from the right-hand side of (32), $\bar{v}_5/\mu_5 = \bar{v}_6/\mu_6 = 6$ is greater than 4.75 calculated from the right-hand side of (32); $\mu_i = 0.1$ and $\bar{v}_3 = \bar{v}_4 = 0.1$, $\bar{v}_5 = \bar{v}_6 = 0.12$ in the distance + angle controller (65), where the ratio $\bar{v}_3/\mu_3 = \bar{v}_4/\mu_4 = 1$ is greater than 0.4135 calculated from the right-hand side of (79), $\bar{v}_5/\mu_5 = \bar{v}_6/\mu_6 = 1.2$ is greater than 0.5469 calculated from the right-hand side of (79).

The trajectories of the agents from their initial positions until they acquire the desired formation are shown in Figs. 9 and 11 for the distance-and-area and distance-and-angle controllers, respectively. The inter-agent distance errors corresponding to the nine edges in E defined in (24), and signed area errors defined in (25), are shown in Fig. 10 for distance-and-area scheme. The inter-agent distance errors corresponding to the nine edges in E defined in (57), and edge angle errors defined in (58), are shown in Fig. 12 for distance-and-angle scheme. We can see all the errors converge to zero finally, and all the agents are on the correct side as desired. A video of the experiments can be viewed at https://youtu.be/MM_Ag7OjfnM.

7. Conclusions

In this work, we introduced a 2D switching formation controller based on the single-integrator model that uses both distance and signed area or edge angle information to ensure that the desired formation shape is reached. The distance + area and distance + angle schemes are alternative approaches to the multi-equilibrium point problem of the distance-only method. The two schemes yield control laws of comparable complexity, and both ensure asymptotic convergence to the desired equilibrium under certain conditions on the control gains and initial relative agent positions. The main difference between the two schemes

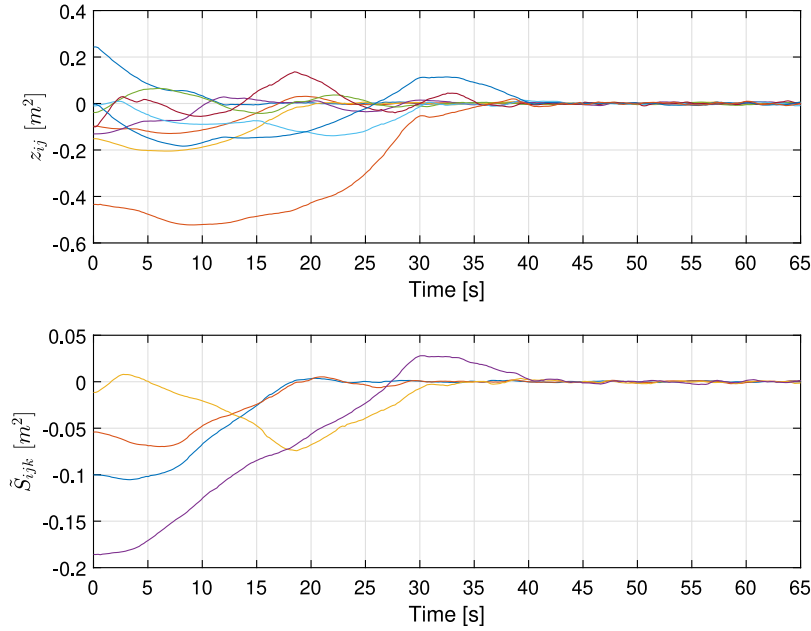


Fig. 10. Distance + area control: Distance errors $z_{ij}(t)$ (top) and signed area errors $\tilde{S}_{ij,k}(t)$ (bottom).

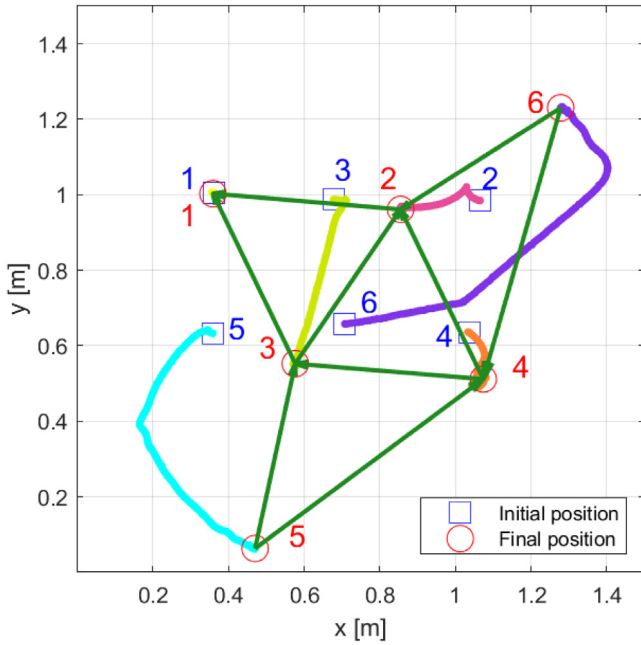


Fig. 11. Distance + angle control: Agent trajectories $p_i(t)$, $i = 1, \dots, 6$ for $t = [0, 80]$ s.

is that no two connected agents can be initially collocated in the angle-based method, whereas this condition only applies to the leader and the first follower in the area-based method. The main contribution of this paper was the introduction of the switching strategy which enabled us to completely eliminate the limitations on the desired formation shape that were present in previous (non-switching) distance + area control schemes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Tairan Liu: Conceptualization, Methodology, Formal analysis, Visualization, Writing - original draft, Writing - review & editing.
Victor Fernández-Kim: Software, Investigation, Formal analysis, Visualization, Funding acquisition, Writing - original draft.
Marcio de Queiroz: Conceptualization, Methodology, Supervision, Project administration, Funding acquisition, Visualization, Writing - original draft, Writing - review & editing.

Appendix A. Proof of Lemma 1

(Proof of $\chi \Rightarrow A$) Let $G = (\mathcal{V}, E)$. If F and $\hat{F}$ are congruent, then

$$\|p_i - p_j\| = \|\hat{p}_i - \hat{p}_j\|, \quad \forall i, j \in \mathcal{V}, i \neq j. \quad (A.1)$$

Since $\chi(p) = \chi(\hat{p})$, then

$$S_{kij}(p) = S_{kij}(\hat{p}), \quad \forall (k, i), (k, j) \in E. \quad (A.2)$$

From (2), we have that

$$\begin{aligned} S_{kij}(p) &= \frac{1}{2} \|p_k - p_i\| \|J(p_k - p_j)\| \cos\left(\frac{\pi}{2} - \alpha_{kij}(p)\right) \\ &= \frac{1}{2} \|p_k - p_i\| \|p_k - p_j\| \sin \alpha_{kij}(p) \end{aligned} \quad (A.3)$$

and $S_{kij}(\hat{p}) = \frac{1}{2} \|\hat{p}_k - \hat{p}_i\| \|\hat{p}_k - \hat{p}_j\| \sin \alpha_{kij}(\hat{p})$. Therefore, given (A.1), we know

$$\sin \alpha_{kij}(p) = \sin \alpha_{kij}(\hat{p}), \quad \forall (k, i), (k, j) \in E. \quad (A.4)$$

From the law of cosines, we obtain

$$\cos \alpha_{kij}(p) = \frac{\|p_k - p_i\|^2 + \|p_k - p_j\|^2 - \|p_j - p_i\|^2}{2 \|p_k - p_i\| \|p_k - p_j\|}$$

and

$$\cos \alpha_{kij}(\hat{p}) = \frac{\|\hat{p}_k - \hat{p}_i\|^2 + \|\hat{p}_k - \hat{p}_j\|^2 - \|\hat{p}_j - \hat{p}_i\|^2}{2 \|\hat{p}_k - \hat{p}_i\| \|\hat{p}_k - \hat{p}_j\|},$$

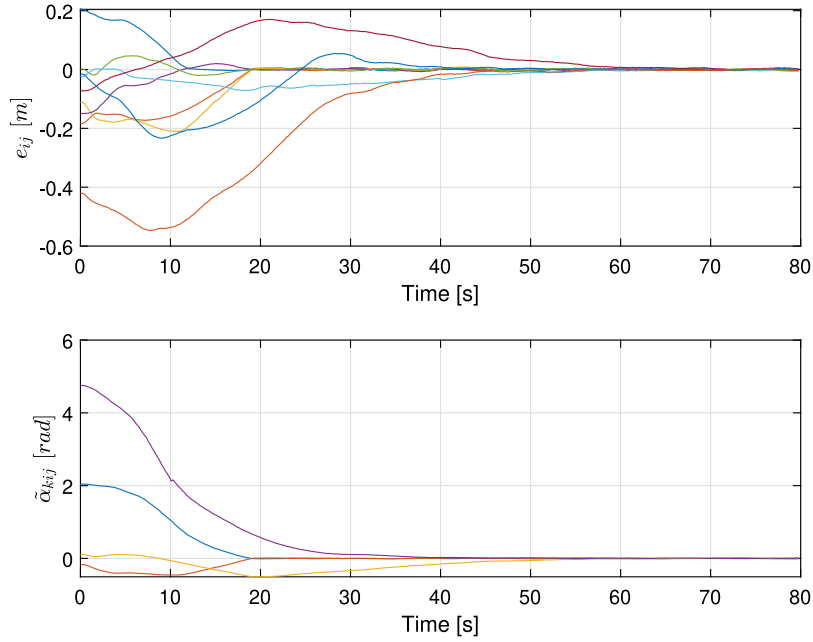


Fig. 12. Distance + angle control: Distance errors $e_{ij}(t)$ (top) and edge angle errors $\tilde{\alpha}_{kij}(t)$ (bottom).

which indicates

$$\cos \alpha_{kij}(p) = \cos \alpha_{kij}(\hat{p}), \quad \forall (k, i), (k, j) \in E. \quad (\text{A.5})$$

Therefore, from (A.4) and (A.5), we know that $\alpha_{kij}(p) = \alpha_{kij}(\hat{p})$, $\forall (k, i), (k, j) \in E$ so $A(p) = A(\hat{p})$.

(Proof of $A \Rightarrow \chi$) Since $A(p) = A(\hat{p})$, then $\alpha_{kij}(p) = \alpha_{kij}(\hat{p})$, $\forall (k, i), (k, j) \in E$. Hence, (A.4) holds and, from (A.1) and (A.3), we know (A.2) holds. Therefore, $\chi(p) = \chi(\hat{p})$.

Appendix B. Proof of Lemma 2

1. (a) $\Leftrightarrow$ (b)

This equivalency was proved in Appendix 1 of [9].

2. (a) $\Leftrightarrow$ (c)

(Proof of $\Rightarrow$) The strong congruency of F and $\hat{F}$ implies that $\|p_i - p_j\| = \|\hat{p}_i - \hat{p}_j\|$, $\forall i, j \in \mathcal{V}$ and $A(p) = A(\hat{p})$ by definition. Thus, since $E \subset \mathcal{V} \times \mathcal{V}$, we have that $\|p_j - p_i\| = \|\hat{p}_j - \hat{p}_i\|$, $\forall (i, j) \in E$, i.e., F and $\hat{F}$ are equivalent.

(Proof of $\Leftarrow$) If the framework has three vertices, i.e., $|\mathcal{V}| = 3$, then framework equivalency and congruency are equivalent, and the conditions for strong congruency are obviously met.

If we add a vertex so that $|\mathcal{V}| = 4$, the resulting framework will have two extra edges and one extra edge angle. Without loss of generality, consider the framework in Fig. B.13(a), where the edge angle α_{314} is given by $\alpha_{314} = \text{mod}\{\alpha_{312} + \alpha_{324}, 2\pi\}$. Since $A(p) = A(\hat{p})$, we know that $\alpha_{312}(p) = \alpha_{312}(\hat{p})$ and $\alpha_{423}(p) = \alpha_{423}(\hat{p})$. Since F and $\hat{F}$ are equivalent, $\|p_2 - p_1\| = \|\hat{p}_2 - \hat{p}_1\|$, $\|p_3 - p_1\| = \|\hat{p}_3 - \hat{p}_1\|$, $\|p_3 - p_2\| = \|\hat{p}_3 - \hat{p}_2\|$, $\|p_4 - p_2\| = \|\hat{p}_4 - \hat{p}_2\|$, and $\|p_4 - p_3\| = \|\hat{p}_4 - \hat{p}_3\|$. From the law of sines, we know that $\alpha_{324}(p) = \alpha_{324}(\hat{p})$; thus, $\alpha_{314}(p) = \alpha_{314}(\hat{p})$. Now, using the law of cosines, we obtain

$$\begin{aligned} & \frac{\|p_3 - p_1\|^2 + \|p_4 - p_3\|^2 + \|p_4 - p_1\|^2}{2 \|p_3 - p_1\| \|p_4 - p_3\|} \\ &= \frac{\|\hat{p}_3 - \hat{p}_1\|^2 + \|\hat{p}_4 - \hat{p}_3\|^2 + \|\hat{p}_4 - \hat{p}_1\|^2}{2 \|\hat{p}_3 - \hat{p}_1\| \|\hat{p}_4 - \hat{p}_3\|}. \end{aligned} \quad (\text{B.1})$$

From (B.1), we have that $\|p_4 - p_1\| = \|\hat{p}_4 - \hat{p}_1\|$, which indicates that F and $\hat{F}$ are strongly congruent for $|\mathcal{V}| = 4$. A similar analysis

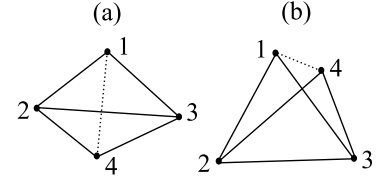


Fig. B.13. Convex and concave quadrilaterals.

can be conducted for all other convex or concave quadrilateral cases, one of which is shown in Fig. B.13 (b). As more vertices and edges are added, we can replicate the above process to conclude that F and $\hat{F}$ are strongly congruent for $|\mathcal{V}| = n$.

Therefore, statements (a) and (c) are equivalent. Since (a) $\Leftrightarrow$ (b) and (a) $\Leftrightarrow$ (c), we know (b) $\Leftrightarrow$ (c).

Appendix C. Proof of Corollary 1

If we let $f(x) := ax^4 + bx^3 + cx^2 + dx + e$, then $f(-x) = ax^4 - bx^3 + cx^2 - dx + e$. Since $a < 0$, $b > 0$, $c < 0$, $d > 0$, and $e < 0$, there are four sign changes in the coefficients of $f(x)$ and no sign changes in the coefficients of $f(-x)$. Therefore, by Lemma 3, $f(x)$ has 0, 2, or 4 positive real roots and $f(-x)$ has no positive roots, which means $f(x)$ has no negative real roots. Furthermore, $f(x)$ has no zero roots since $e \neq 0$. Therefore, the roots of $f(x) = 0$ are either complex or positive real.

Next, we verify the requirements for the coefficients to have the indicated signs.

- $a < 0$ requires $\gamma \neq 2$ and (11) to be positive.
- $b > 0$ requires $\gamma > 2$.
- $c < 0$ requires

$$\gamma > 0 \quad \text{if} \quad \frac{\delta_2^2 + \delta_3^2}{\delta_1^2} < \frac{9}{2}$$

or

$$\gamma > \frac{3}{2} + \frac{\delta_2^2 + \delta_3^2}{\delta_1^2} + \sqrt{\left(\frac{\delta_2^2 + \delta_3^2}{\delta_1^2} - \frac{9}{2}\right)\left(\frac{\delta_2^2 + \delta_3^2}{\delta_1^2} - \frac{1}{2}\right)} \quad \text{if} \quad \frac{\delta_2^2 + \delta_3^2}{\delta_1^2} \geq \frac{9}{2}.$$

- $d > 0$ requires $\gamma > \frac{3}{2} + \frac{d_{ki}^2 + d_{kj}^2}{d_{ji}^2}$ and (11) to be positive.
- $e < 0$ requires $\gamma > 0$ and (11) to be positive.

If θ is the angle between edges δ_1 and δ_2 of the triangle, then

$$\cos \theta = \frac{\delta_1^2 + \delta_2^2 - \delta_3^2}{2\delta_1\delta_2}$$

and

$$\sin^2 \theta = 1 - \cos^2 \theta = \frac{2\delta_1^2\delta_2^2 + 2\delta_1^2\delta_3^2 + 2\delta_2^2\delta_3^2 - \delta_1^4 - \delta_2^4 - \delta_3^4}{4\delta_1^2\delta_2^2} = \frac{\rho}{4\delta_1^2\delta_2^2}.$$

Since $\theta \neq 0, \pi$, then $\sin^2 \theta > 0$ and (11) is positive.

Since

$$\begin{aligned} \frac{3}{2} + \frac{\delta_2^2 + \delta_3^2}{\delta_1^2} &\geq \frac{3}{2} + \frac{\delta_2^2 + \delta_3^2 + 2\delta_2\delta_3}{2\delta_1^2} \\ &= \frac{3}{2} + \frac{(\delta_2 + \delta_3)^2}{2\delta_1^2} > \frac{3}{2} + \frac{\delta_1^2}{2\delta_1^2} = 2, \end{aligned} \quad (\text{C.1})$$

we obtain the overall requirement on γ shown in (12).

Appendix D. Proof of Lemma 9

To facilitate the analysis of (78), we consider the following: (1) We assume the angle α_{kij}^* is such that $0 < \alpha_{kij}^* < \pi$; (2) We divide the 2D plane where the agents are deployed into regions based on the values of the variables e_{ki} , e_{kj} , and α_{kij} as shown in Fig. D.14; and (3) We use the fact that (78) is not affected by translations and rotations of the triangular framework since it is a function of the relative position of agents and the edge angle. Thus, projections of (78) can be made onto orthogonal non-trivial vectors without changing its solution. Specifically, if $\tilde{p}_{ki} \neq 0$, then (78) can be projected onto vectors in the directions of $\tilde{p}_{ki}$ and $J^\top \tilde{p}_{ki}$ to yield

$$w_{ki} \|\tilde{p}_{ki}\|^2 + w_{kj} \|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\| \cos \alpha_{kij} = \frac{v_k}{\mu_k} \tilde{\alpha}_{kij} \frac{\|\tilde{p}_{kj}\| \sin \alpha_{kij}}{\|\tilde{p}_{kj}\|} \quad (\text{D.1})$$

$$w_{kj} \|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\|^2 \sin \alpha_{kij} = \frac{v_k}{\mu_k} \tilde{\alpha}_{kij} (\|\tilde{p}_{kj}\| - \|\tilde{p}_{ki}\| \cos \alpha_{kij}), \quad (\text{D.2})$$

respectively. Since $\tilde{p}_{ki}$ and $J^\top \tilde{p}_{ki}$ are orthogonal, if (D.1) or (D.2) do not hold within a given region, then (78) will not have a solution in that region. Alternatively, if $\tilde{p}_{kj} \neq 0$, projections onto $\tilde{p}_{kj}$ and $J^\top \tilde{p}_{kj}$ give

$$w_{kj} \|\tilde{p}_{kj}\|^2 + w_{ki} \|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\| \cos \alpha_{kij} = \frac{v_k}{\mu_k} \tilde{\alpha}_{kij} \frac{\|\tilde{p}_{ki}\| \sin \alpha_{kij}}{\|\tilde{p}_{ki}\|} \quad (\text{D.3})$$

$$w_{ki} \|\tilde{p}_{ki}\|^2 \|\tilde{p}_{kj}\| \sin \alpha_{kij} = \frac{v_k}{\mu_k} \tilde{\alpha}_{kij} (\|\tilde{p}_{ki}\| - \|\tilde{p}_{kj}\| \cos \alpha_{kij}), \quad (\text{D.4})$$

respectively. Finally, if $\frac{\tilde{p}_{ki}}{\|\tilde{p}_{ki}\|} + \frac{\tilde{p}_{kj}}{\|\tilde{p}_{kj}\|} \neq 0$, projections can also be made onto $\frac{\tilde{p}_{ki}}{\|\tilde{p}_{ki}\|} + \frac{\tilde{p}_{kj}}{\|\tilde{p}_{kj}\|}$ and $J^\top (\frac{\tilde{p}_{ki}}{\|\tilde{p}_{ki}\|} + \frac{\tilde{p}_{kj}}{\|\tilde{p}_{kj}\|})$:

$$(w_{ki} \|\tilde{p}_{ki}\| + w_{kj} \|\tilde{p}_{kj}\|)(1 + \cos \alpha_{kij})$$

$$= \frac{v_k}{\mu_k} \tilde{\alpha}_{kij} \frac{\|\tilde{p}_{ki}\| + \|\tilde{p}_{kj}\|}{\|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\|} \sin \alpha_{kij} \quad (\text{D.5})$$

$$\begin{aligned} &(w_{kj} \|\tilde{p}_{kj}\| - w_{ki} \|\tilde{p}_{ki}\|) \sin \alpha_{kij} \\ &= \frac{v_k}{\mu_k} \tilde{\alpha}_{kij} \frac{\|\tilde{p}_{kj}\| - \|\tilde{p}_{ki}\|}{\|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\|} (1 + \cos \alpha_{kij}). \end{aligned} \quad (\text{D.6})$$

Note that $\alpha_{kij} = \pi$ implies p_k is between p_i and p_j , which will cause $\frac{\tilde{p}_{ki}}{\|\tilde{p}_{ki}\|} + \frac{\tilde{p}_{kj}}{\|\tilde{p}_{kj}\|} = 0$. For this case, projections (D.5) and (D.6) cannot be used.

In the following, one of the above projections will be used in the analysis of the different regions in Fig. D.14.

Region 1 ($0 < \alpha_{kij} < \pi$): In this region, the + subsystem of the switched system is active, i.e., $q = +$. Since $\sin \alpha_{kij} > 0$ in this region, we have from (D.2) with $v_k = 0$ that $w_{kj} = 0$, i.e., $\|\tilde{p}_{kj}\| = d_{kj}$. Substituting this result into (D.1) with $v_k = 0$ gives $w_{ki} = 0$, i.e., $\|\tilde{p}_{ki}\| = d_{ki}$. Furthermore, we obtain from the law of cosines that

$$\cos \alpha_{kij} = \frac{\|\tilde{p}_{ki}\|^2 + \|\tilde{p}_{kj}\|^2 - d_{ji}^2}{2 \|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\|} = \frac{d_{ki}^2 + d_{kj}^2 - d_{ji}^2}{2d_{ki}d_{kj}} = \cos \alpha_{kij}^*. \quad (\text{D.7})$$

That is, when $0 < \alpha_{kij} < \pi$, the only solution to (78) corresponds to $[e_{ki}, e_{kj}, \tilde{\alpha}_{kij}] = 0$.

Region 2 ($\pi < \alpha_{kij} < 2\pi$, $e_{ki} < 0$, and $e_{kj} < 0$): In this region (and all subsequent ones), $q = -$ and $v_k = \tilde{v}_k$.

If $\|\tilde{p}_{ki}\| - \|\tilde{p}_{kj}\| \cos \alpha_{kij} \leq 0$, the left-hand side (LHS) of (D.4) is positive while the right-hand side (RHS) is negative. Therefore, no solution exists. If $\|\tilde{p}_{kj}\| - \|\tilde{p}_{ki}\| \cos \alpha_{kij} \leq 0$, the LHS of (D.2) is positive while the RHS is negative, so again no solution exists.

If $\|\tilde{p}_{ki}\| - \|\tilde{p}_{kj}\| \cos \alpha_{kij} > 0$ and $\|\tilde{p}_{kj}\| - \|\tilde{p}_{ki}\| \cos \alpha_{kij} > 0$, then

$$\|\tilde{p}_{ki}\|^2 + d_{ji}^2 > \|\tilde{p}_{kj}\|^2 \quad \text{and} \quad \|\tilde{p}_{kj}\|^2 + d_{ji}^2 > \|\tilde{p}_{ki}\|^2. \quad (\text{D.8})$$

If we rewrite (D.5) as $g_r = \frac{v_k}{\mu_k} g_\alpha$ where

$$g_r(\|\tilde{p}_{ki}\|, \|\tilde{p}_{kj}\|) := \frac{w_{ki} \|\tilde{p}_{ki}\| + w_{kj} \|\tilde{p}_{kj}\|}{\frac{1}{\|\tilde{p}_{ki}\|} + \frac{1}{\|\tilde{p}_{kj}\|}} \quad (\text{D.9})$$

and

$$g_\alpha(\alpha_{kij}) := \tilde{\alpha}_{kij} \frac{\sin \alpha_{kij}}{1 + \cos \alpha_{kij}}, \quad (\text{D.10})$$

we have that

$$\begin{aligned} g_r &= \frac{\frac{\|\tilde{p}_{ki}\|^2 \|\tilde{p}_{kj}\|}{d_{ki}^2} + \frac{\|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\|^2}{d_{kj}^2}}{\|\tilde{p}_{ki}\| + \|\tilde{p}_{kj}\|} - 1 > -1 \\ g_r &< \frac{\frac{d_{ki}^2 \|\tilde{p}_{kj}\|}{d_{ki}^2} + \frac{\|\tilde{p}_{ki}\| d_{kj}^2}{d_{kj}^2}}{\|\tilde{p}_{ki}\| + \|\tilde{p}_{kj}\|} - 1 = 0, \end{aligned} \quad (\text{D.11})$$

indicating that

$$-1 < g_r < 0. \quad (\text{D.12})$$

Now, from the law of cosines, we obtain

$$\cos \alpha_{kij} = \frac{\|\tilde{p}_{ki}\|^2 + \|\tilde{p}_{kj}\|^2 - d_{ji}^2}{2 \|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\|} =: \psi(\|\tilde{p}_{ki}\|, \|\tilde{p}_{kj}\|) \quad (\text{D.13})$$

where

$$\frac{\partial \psi}{\partial \|\tilde{p}_{ki}\|} = \frac{\|\tilde{p}_{ki}\|^2 - \|\tilde{p}_{kj}\|^2 + d_{ji}^2}{2 \|\tilde{p}_{ki}\|^2 \|\tilde{p}_{kj}\|} \quad (\text{D.14})$$

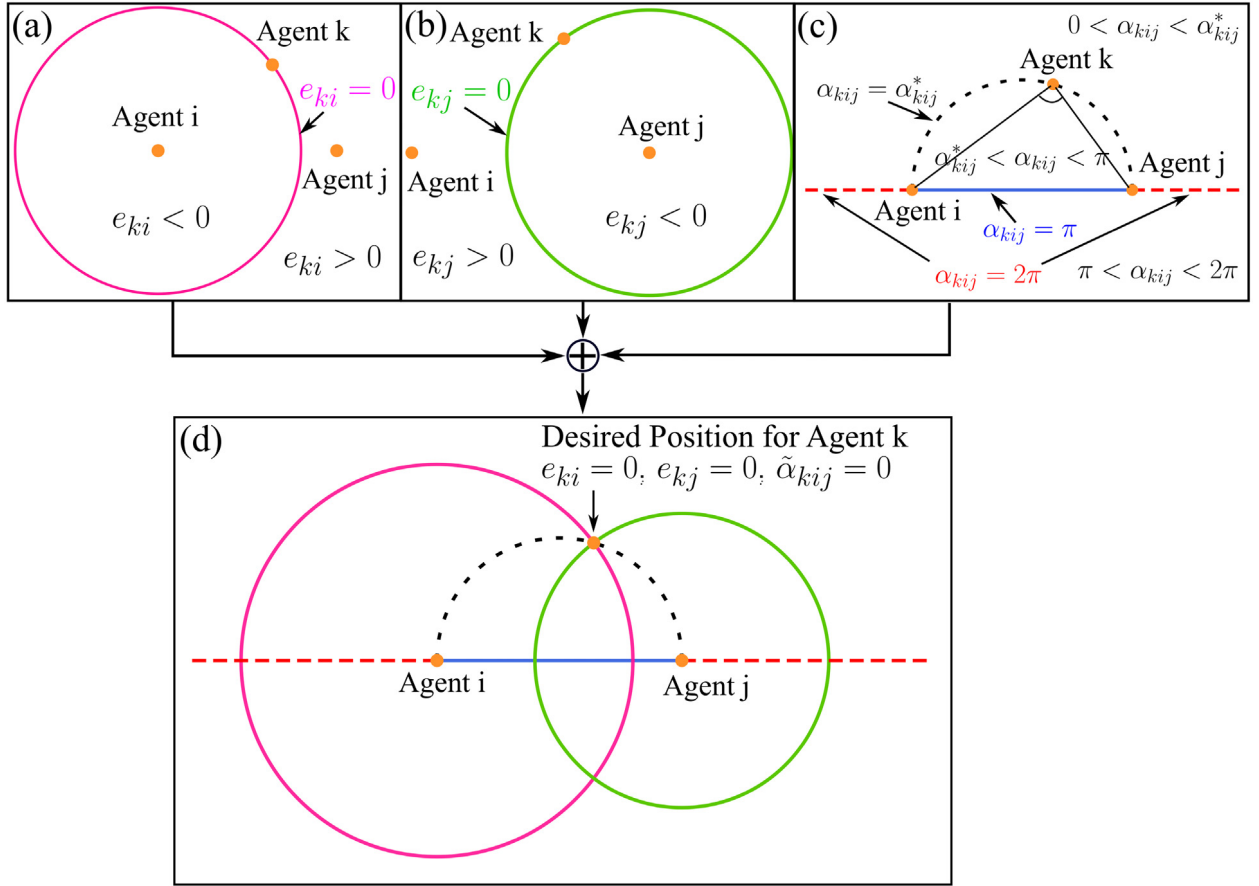


Fig. D.14. The 2D Euclidean space partitioned by the values of e_{ki} , e_{kj} , and α_{kij} . Panel (a): The pink circle is the locus of positions for agent k satisfying $e_{ki} = 0$; the region inside (outside) the circle is the locus for $e_{ki} < 0$ ($e_{ki} > 0$). Panel (b): Similar loci exist with respect to the green circle for values of e_{kj} . Panel (c): The black dashed arc is the locus of positions for agent k satisfying $\alpha_{kij} = \alpha_{kij}^*$ (i.e., $\tilde{\alpha}_{kij} = 0$); the red dashed line is the locus for $\alpha_{kij} = 2\pi$; the blue solid line is the locus for $\alpha_{kij} = \pi$; the region bounded by the black dashed arc and blue solid line corresponds to $\alpha_{kij}^* < \alpha_{kij} < \pi$; the region above the arc and the red dashed line corresponds to $0 < \alpha_{kij} < \alpha_{kij}^*$; the region below the red dashed line and blue solid line is associated with $\pi < \alpha_{kij} < 2\pi$. Panel (d) shows that the intersection of the three regions, $e_{ki} = e_{kj} = \tilde{\alpha}_{kij} = 0$, uniquely defines the desired position for agent k . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$\frac{\partial \psi}{\partial \|\tilde{p}_{kj}\|} = \frac{\|\tilde{p}_{kj}\|^2 - \|\tilde{p}_{ki}\|^2 + d_{ji}^2}{2 \|\tilde{p}_{ki}\| \|\tilde{p}_{kj}\|^2}. \quad (\text{D.15})$$

It follows from (D.8) that (D.14) and (D.15) are both positive. Therefore, $\psi(\|\tilde{p}_{ki}\|, \|\tilde{p}_{kj}\|) < \psi(d_{ki}, d_{kj})$, i.e., $\cos \alpha_{kij} < \cos \alpha_{kij}^* = \cos(2\pi - \alpha_{kij}^*)$, and $\pi < \alpha_{kij} < 2\pi - \alpha_{kij}^*$.

From (D.10), we obtain

$$\frac{dg_\alpha}{d\alpha_{kij}} = \frac{\alpha_{kij} - \alpha_{kij}^* + \sin \alpha_{kij}}{1 + \cos \alpha_{kij}}. \quad (\text{D.16})$$

Since

$$\frac{d(\alpha_{kij} - \alpha_{kij}^* + \sin \alpha_{kij})}{d\alpha_{kij}} = 1 + \cos \alpha_{kij} > 0$$

when $\pi < \alpha_{kij} < 2\pi - \alpha_{kij}^*$, we know $\alpha_{kij} - \alpha_{kij}^* + \sin \alpha_{kij} > \pi - \alpha_{kij}^* + \sin \pi > 0$. Thus, (D.16) is positive for $\pi < \alpha_{kij} < 2\pi - \alpha_{kij}^*$, and

$$g_\alpha(\alpha_{kij}) < g_\alpha(2\pi - \alpha_{kij}^*) = \frac{-(2\pi - 2\alpha_{kij}^*) \sin \alpha_{kij}^*}{1 + \cos \alpha_{kij}^*} < 0. \quad (\text{D.17})$$

From (D.12) and (D.17), we obtain

$$\frac{g_r}{g_\alpha} < \frac{1 + \cos \alpha_{kij}^*}{(2\pi - 2\alpha_{kij}^*) \sin \alpha_{kij}^*}. \quad (\text{D.18})$$

Thus, if the gain ratio is chosen according to (79), then $g_r = \frac{v_k}{\mu_k} g_\alpha$ and its equivalent form (D.5) will not hold, so (78) has no solution in this region.

Region 3 ($\pi < \alpha_{kij} < 2\pi$, $e_{ki} > 0$, and $e_{kj} < 0$): If $\|\tilde{p}_{ki}\| - \|\tilde{p}_{kj}\| \cos \alpha_{kij} > 0$, the LHS of (D.4) is negative while the RHS is positive (no solution). If $\|\tilde{p}_{ki}\| - \|\tilde{p}_{kj}\| \cos \alpha_{kij} \leq 0$, it follows from (D.13) that

$$\|\tilde{p}_{ki}\|^2 + d_{ji}^2 \leq \|\tilde{p}_{kj}\|^2, \quad (\text{D.19})$$

and

$$\|\tilde{p}_{kj}\|^2 + d_{ji}^2 > \|\tilde{p}_{ki}\|^2. \quad (\text{D.20})$$

From (D.19), (D.20), (D.14), and (D.15), we know that $\frac{\partial \psi}{\partial \|\tilde{p}_{ki}\|} \leq 0$

and $\frac{\partial \psi}{\partial \|\tilde{p}_{kj}\|} > 0$ so $\psi(\|\tilde{p}_{ki}\|, \|\tilde{p}_{kj}\|) < \psi(d_{ki}, d_{kj}) = \cos \alpha_{kij}^*$. Since $\pi < \alpha_{kij} < 2\pi$, we know that $\pi < \alpha_{kij} < 2\pi - \alpha_{kij}^*$. Similar to Region 2, we also have $g_r > -1$ and (D.17); hence, (D.18) holds. If $\bar{v}_k/\mu_k$ satisfies (79), then again $g_r = \frac{\bar{v}_k}{\mu_k} g_\alpha$ does not hold and (78) has no solution.

Remaining Regions. For the other regions in Fig. D.14, one can verify the non-existence of solution by checking the signs of the LHS and RHS of (D.1), (D.2), (D.3), (D.4), (D.5), or (D.6).

In summary, the unique solution to (78) exists when $\|\tilde{p}_{ki}\| = d_{ki}$ and $\|\tilde{p}_{kj}\| = d_{kj}$, $\alpha_{kij} = \alpha_{kij}^*$, which corresponds to the desired equilibrium point $[e_{ki}, e_{kj}, \tilde{\alpha}_{kij}] = 0$.

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